https://www.youtube.com/watch?v=frLwRLS_ZR0

Ray Casting

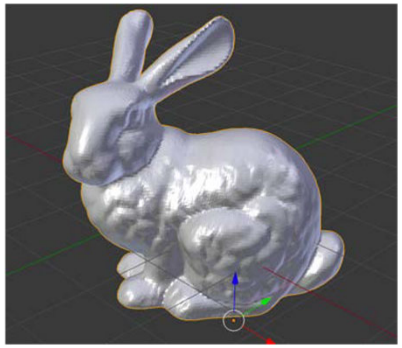
Some Slides/Images adapted from Marschner and Shirley

Topics

- Rasterization vs. Ray Casting
- The Ray Casting Algorithm
- Introduction to Rays
- The Camera
- Ray-Object Intersection
    - Ray-Plane Intersection
    - Ray-Sphere Intersection
    - Ray-Triangle Intersection

Photography

Input:



Objects

Lights

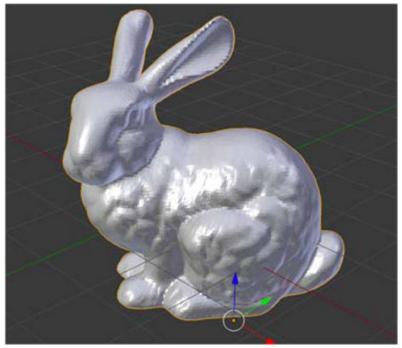
Camera

Output:

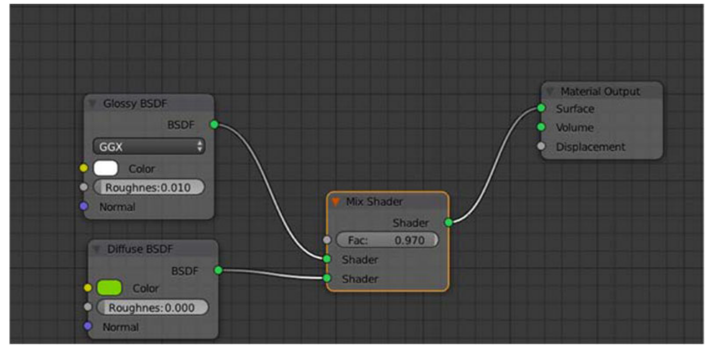
$I\left(x, y\right)$

Rendering

Input:

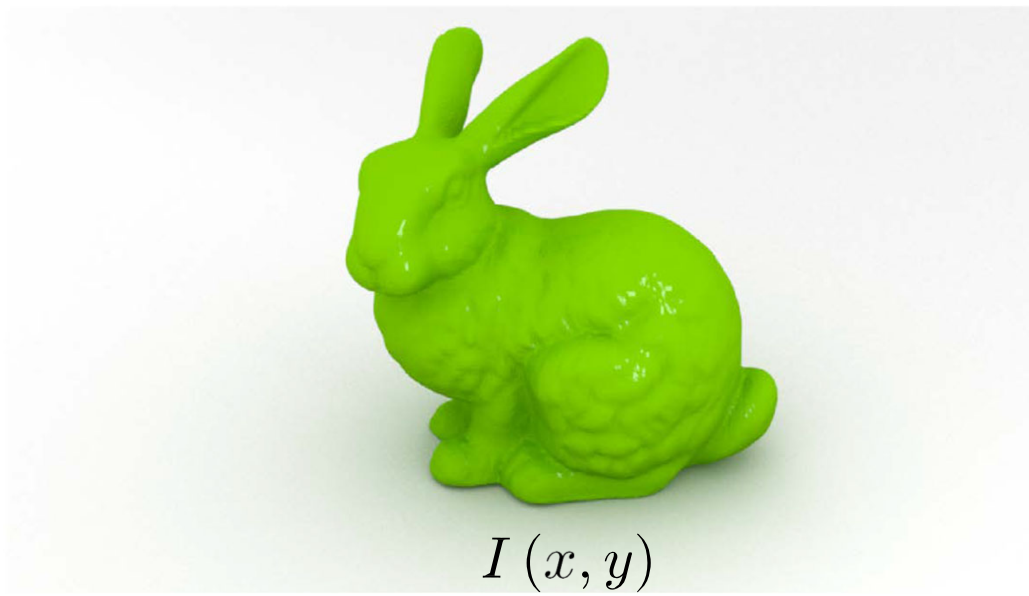


Objects

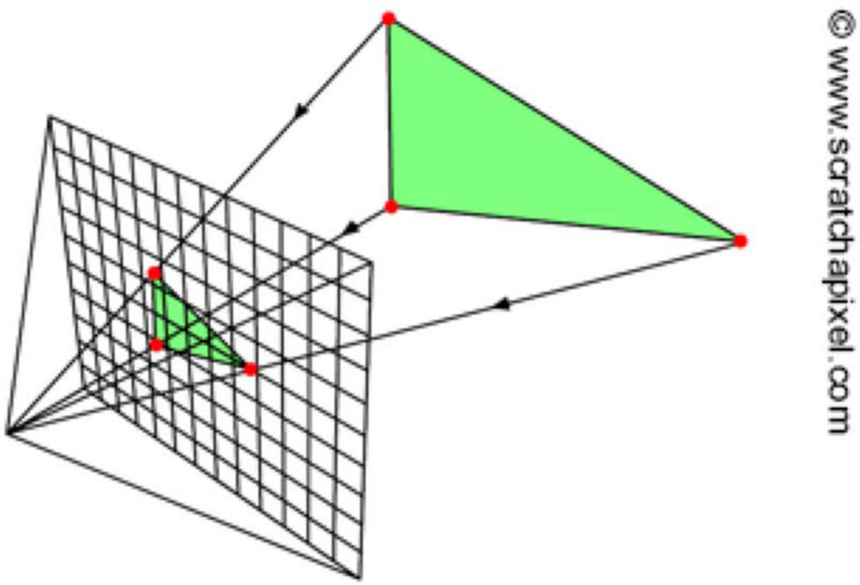


Materials

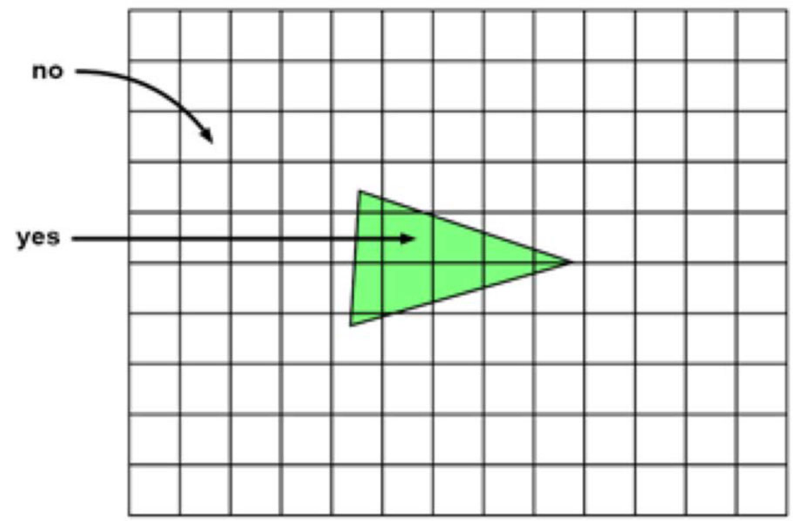
Output:



Rasterization



1. Project Vertices to Image Plane



2. Turn on pixels inside triangle

https://www.scratchapixel.com/lessons/3d-basic-rendering/rasterization-practical-implementation

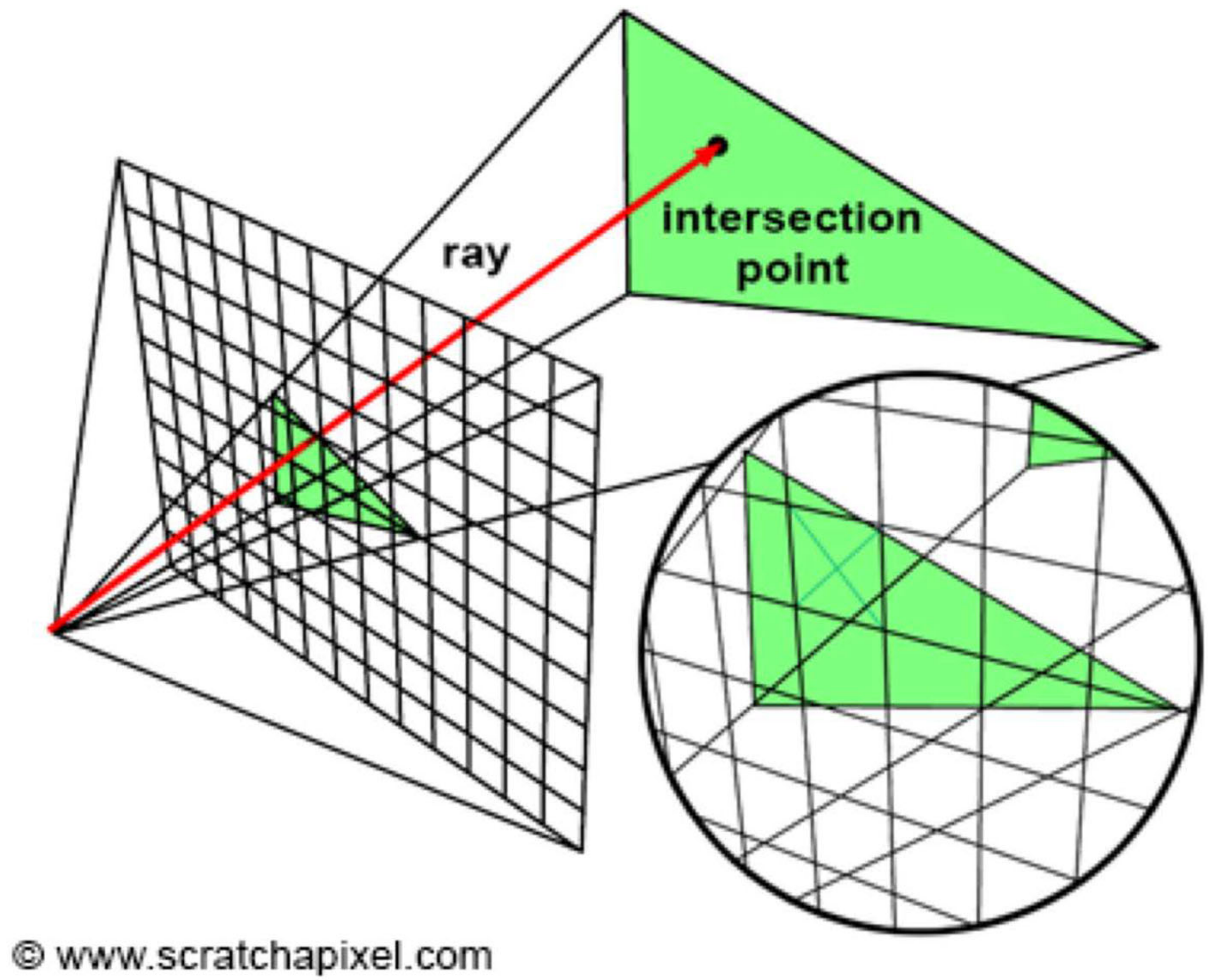
Fortnite | Epic Games

Rasterization

```
for each scene object {
    for each image pixel{
        if (object affects pixel) {
            do something
        }
    }
}
```

operations can be done quickly on the GPU!

Ray Casting



https://www.scratchapixel.com/lessons/3d-basic-rendering/rasterization-practical-implementation

Ray Casting

```
for each image pixel {
    generate a ray
    for each scene object {
        if (ray intersects object) {
            set pixel color
        }
    }
}
```

Ray Casting vs. Rasterization

Ray Casting

```
for each image pixel {
    for each scene object {
        ...
    }
}
```

Rasterization

```
for each scene object {
    for each image pixel {
        ...
    }
}
```

Basic Components of Ray Casting

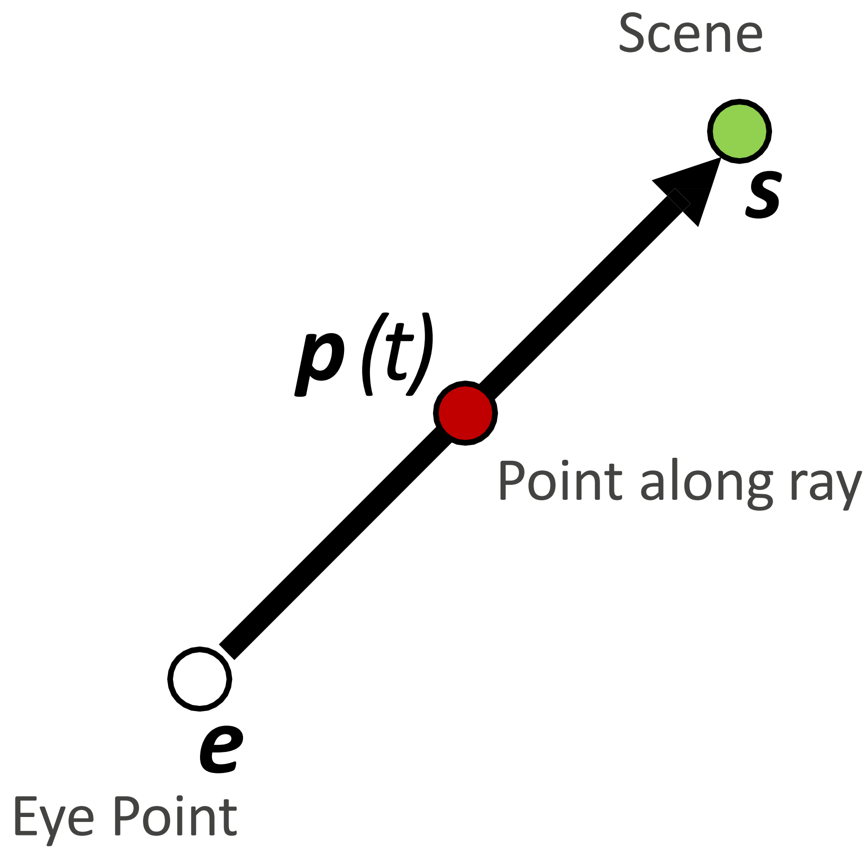
- Ray
- Camera
- Intersection Tests

Ray

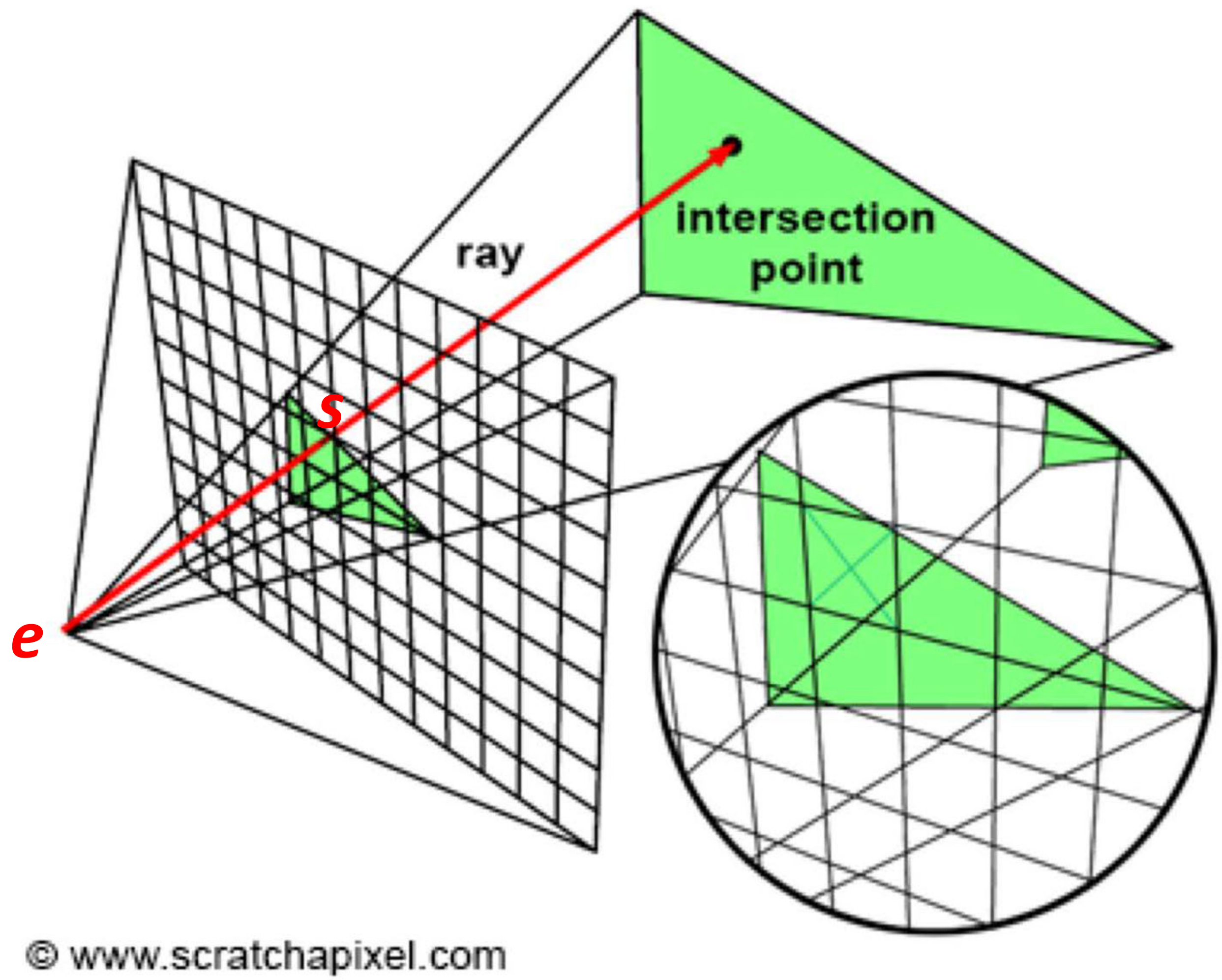
$$\boldsymbol{p}(t) = \boldsymbol{e} + t(\boldsymbol{s} - \boldsymbol{e})$$

$$\boldsymbol{p}(0) = \boldsymbol{e}$$

$$\boldsymbol{p}(1) = \boldsymbol{s}$$

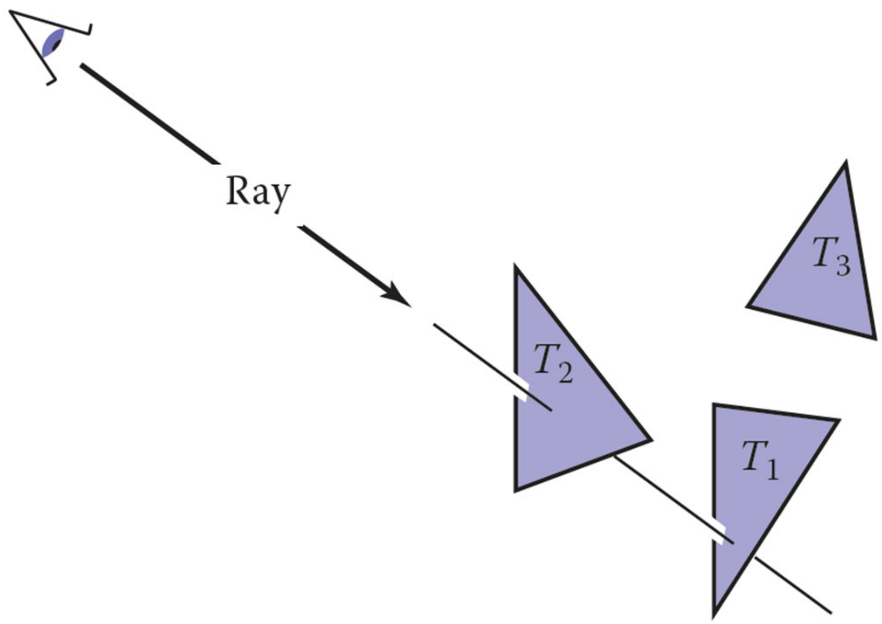


Ray Casting



https://www.scratchapixel.com/lessons/3d-basic-rendering/rasterization-practical-implementation

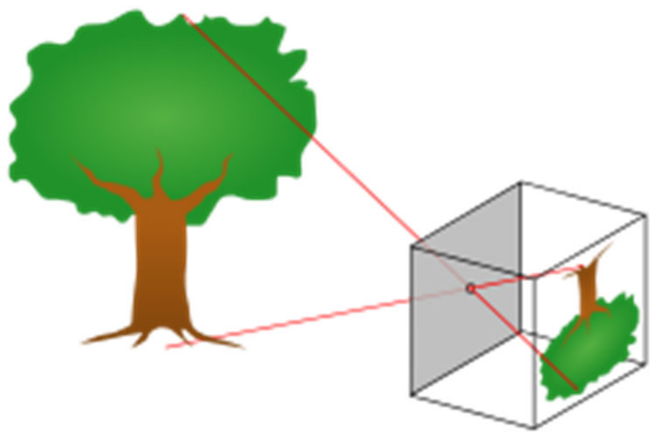
Ray Casting



Basic Components of Ray Casting

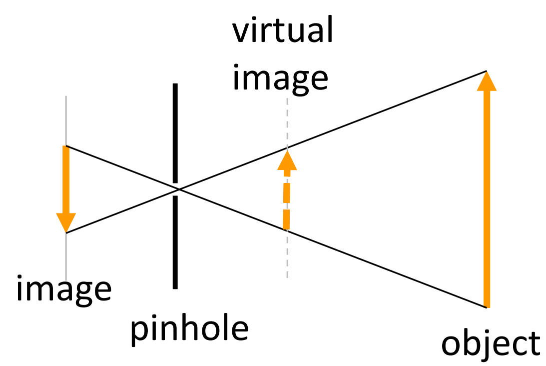
- Ray
- Camera
- Intersection Tests

Camera model: camera obscura

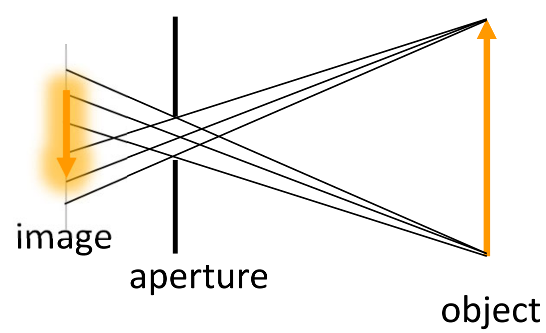


Camera model

Ideal pinhole camera

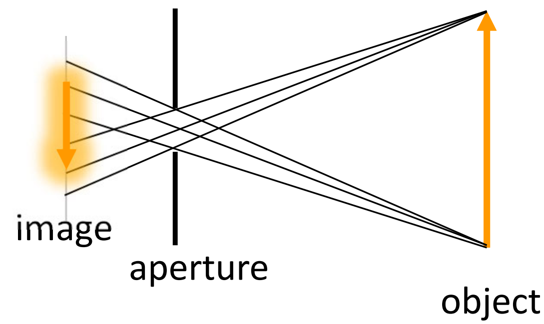


Real pinhole camera

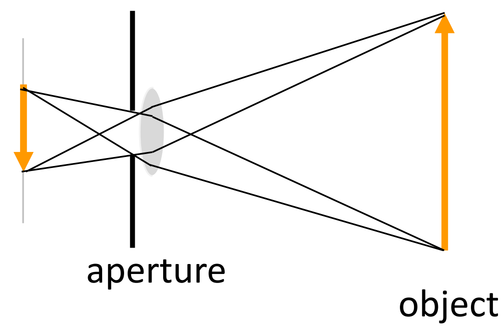


Camera model

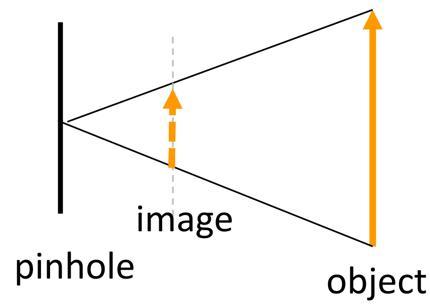
Real pinhole camera



Camera with a lens



Virtual camera

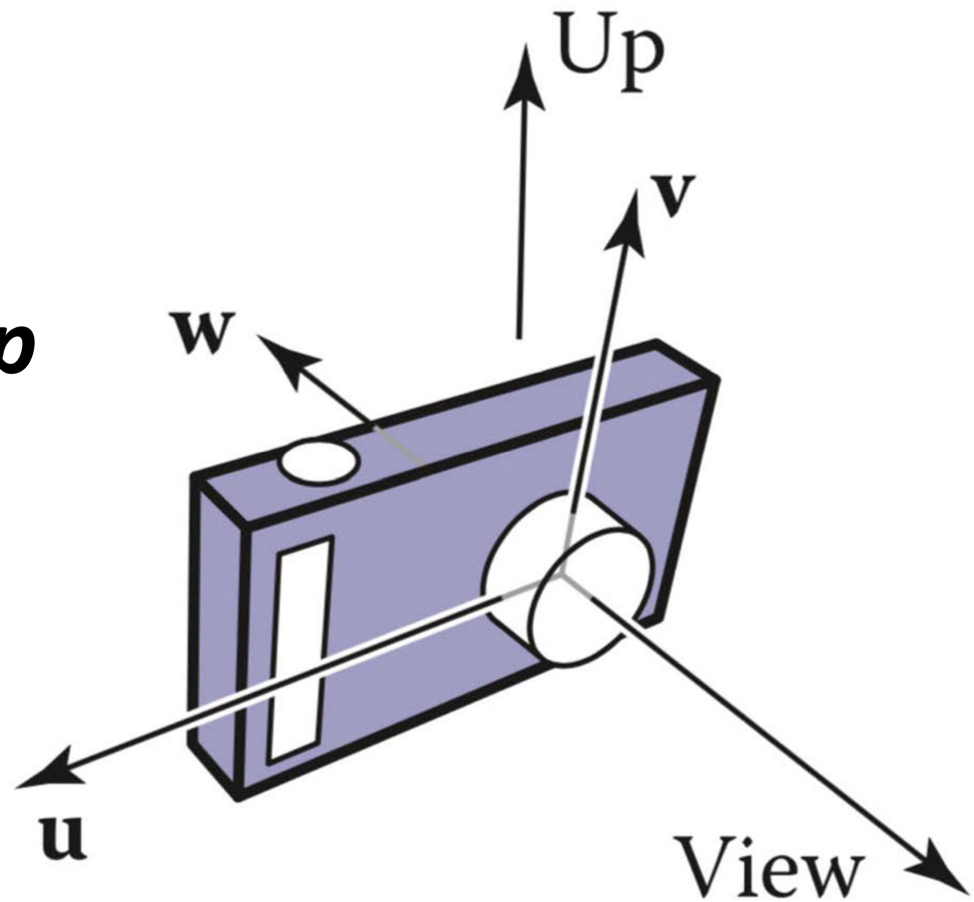


Camera

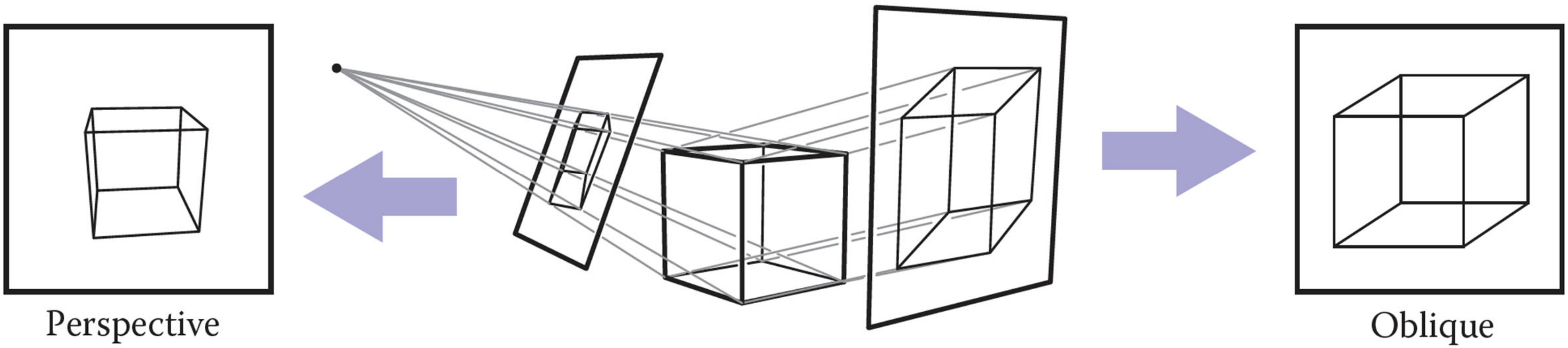
$\boldsymbol{w} = -\boldsymbol{view}$

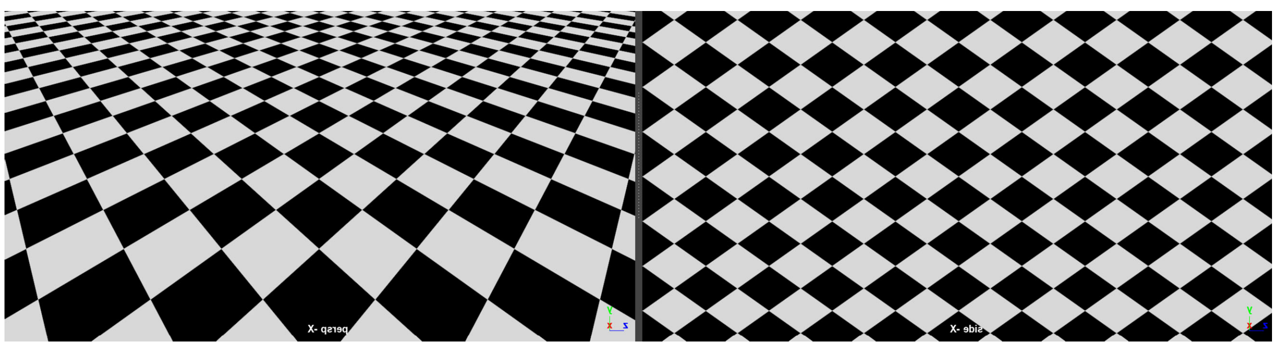
$\boldsymbol{u} = \boldsymbol{view} \times \boldsymbol{up}$

$\boldsymbol{v} = \boldsymbol{w} \times \boldsymbol{u}$

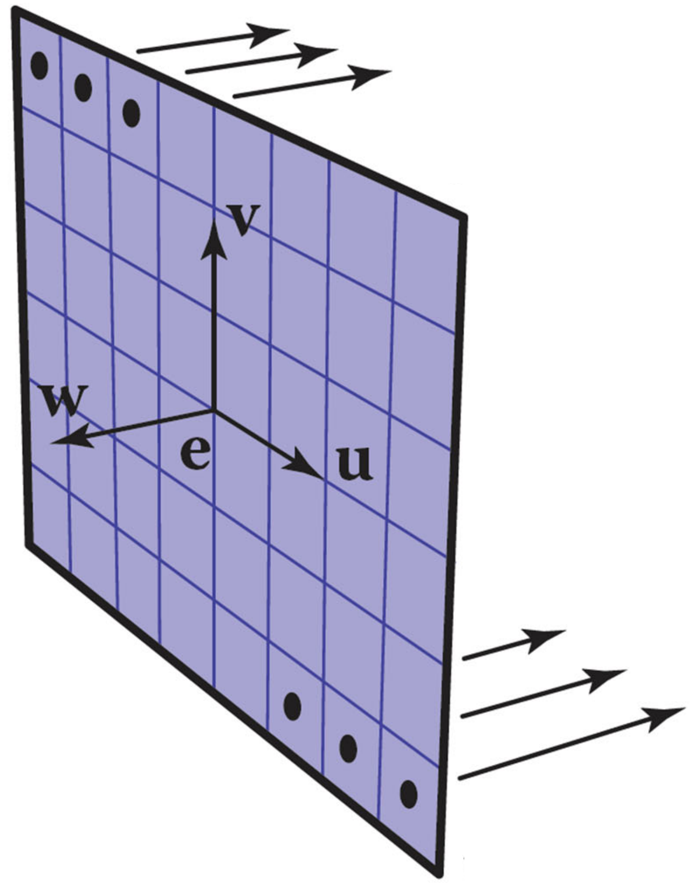


Orthographic v Perspective Projection

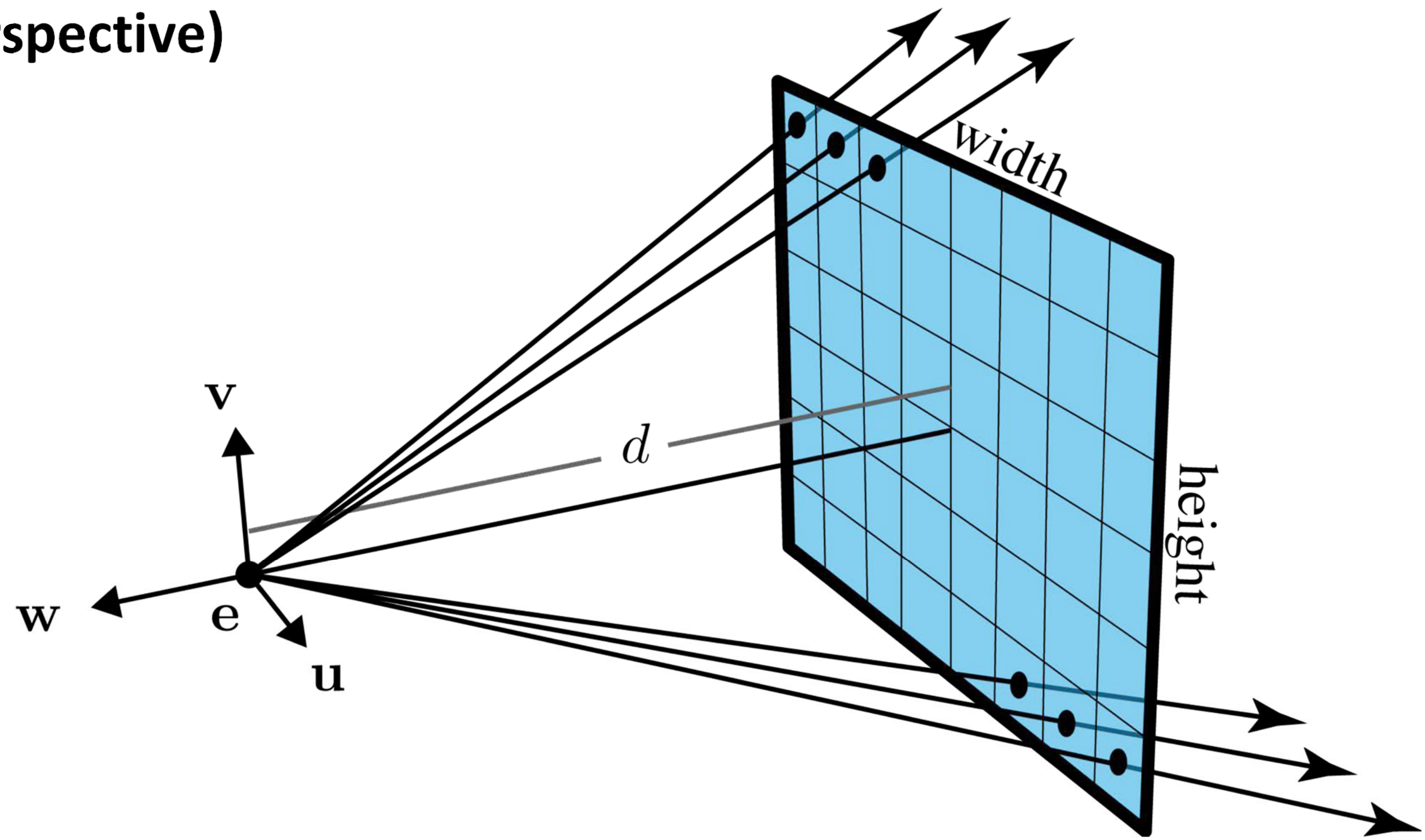




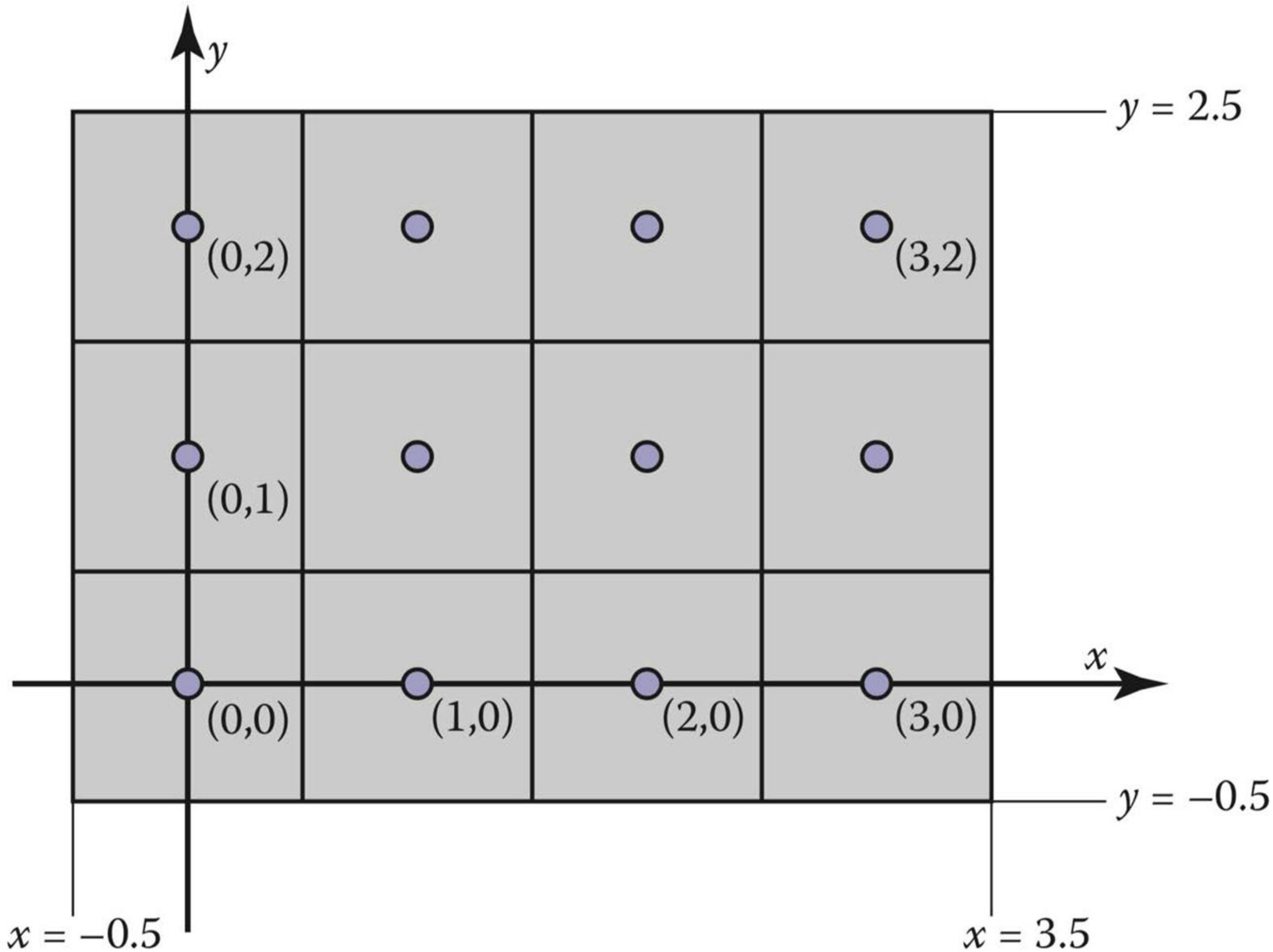
Generating Rays (Orthographic)

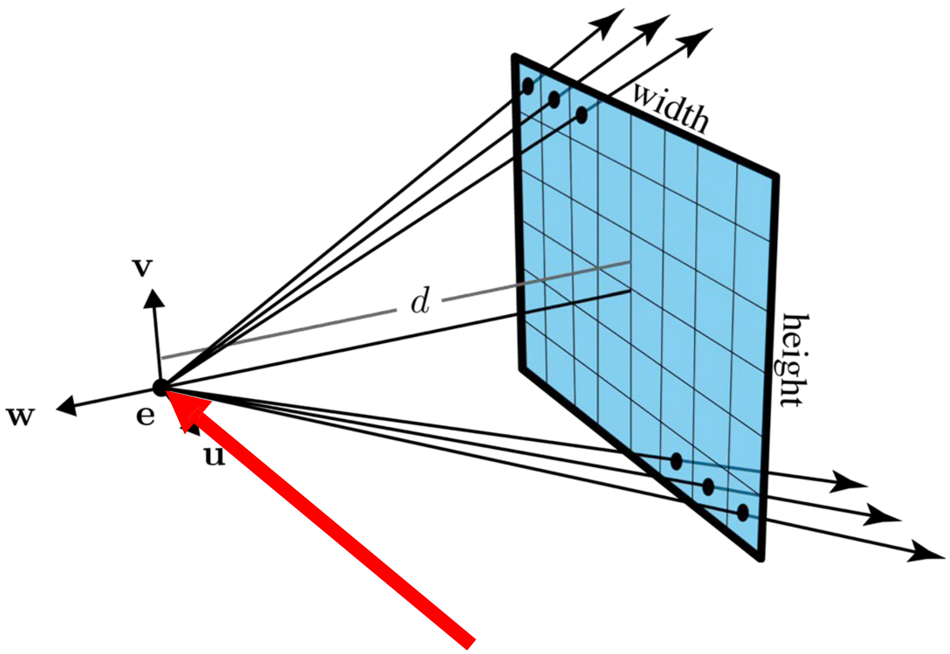


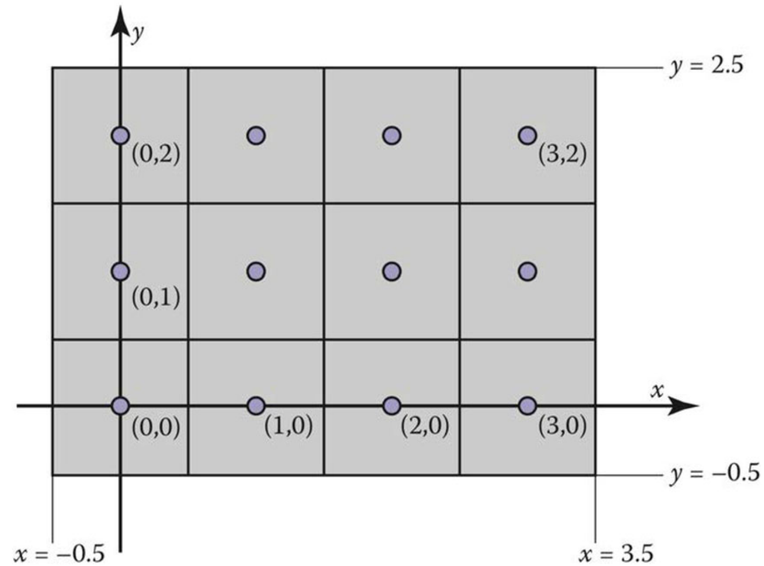
Generating Rays (Perspective)



Recall: Standard Pixel Coordinate System

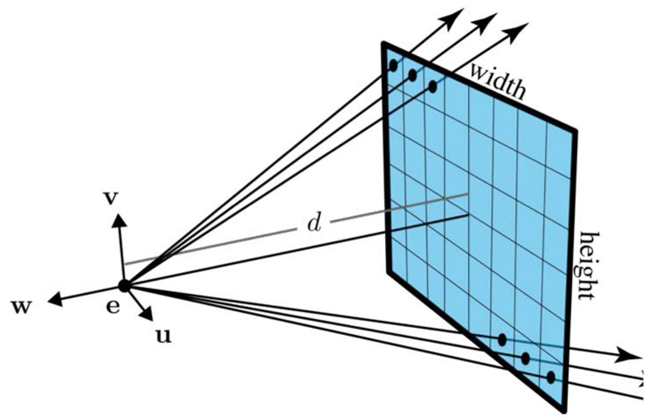




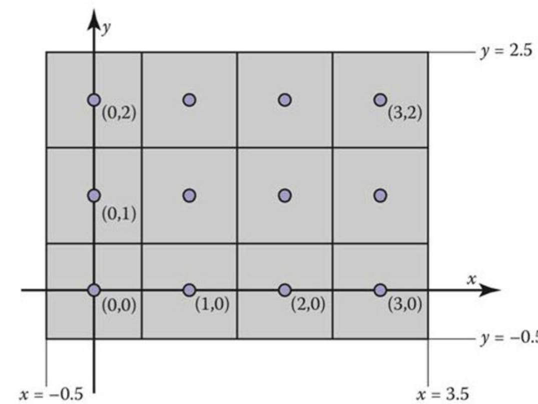


Origin of camera frame (the eye)

What are the *(u,v)* coordinates for pixel (i, j) in the camera frame?



Camera space



Pixel space

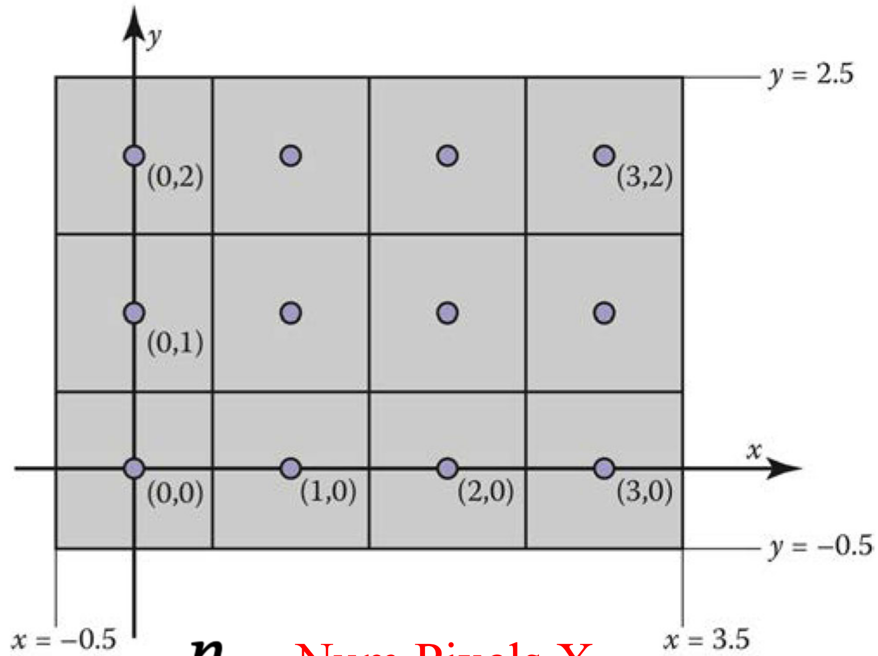
Bottom Left Corner (i, j) : ?

Top Right Corner (i, j): ?

Bottom Left Corner (u, v) : ?

Top Right Corner (u, v) : ?

n_y Num Pixels Y



n_x Num Pixels X

Bottom Left Corner (i, j) : $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

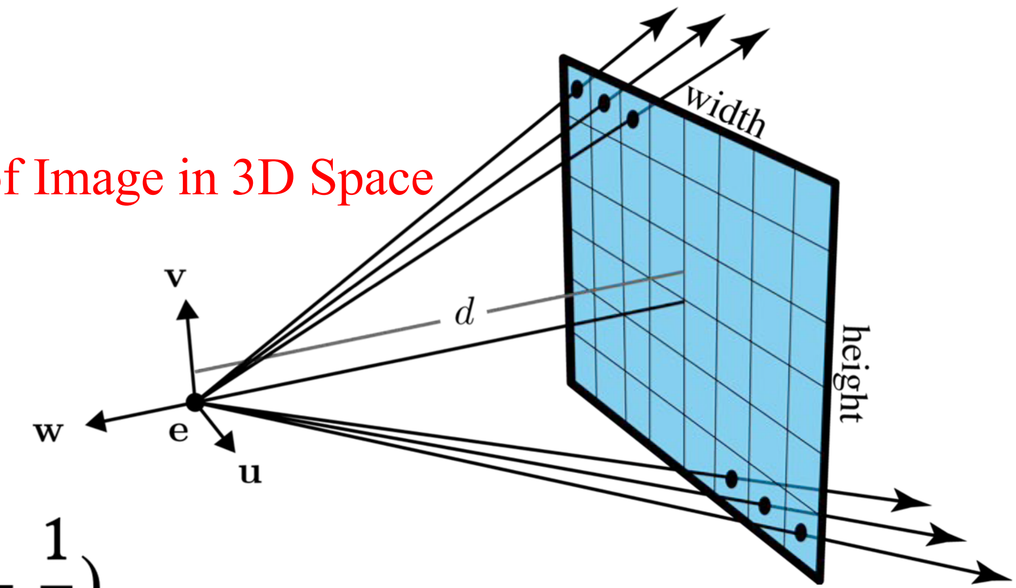
Top Right Corner (i, j) : $\left(n_x - \frac{1}{2}, n_y - \frac{1}{2}\right)$

Bottom Left Corner (u, v) :

Top Right Corner (u, v) :

Physical Width of Image in 3D Space

Physical Height of Image in 3D Space



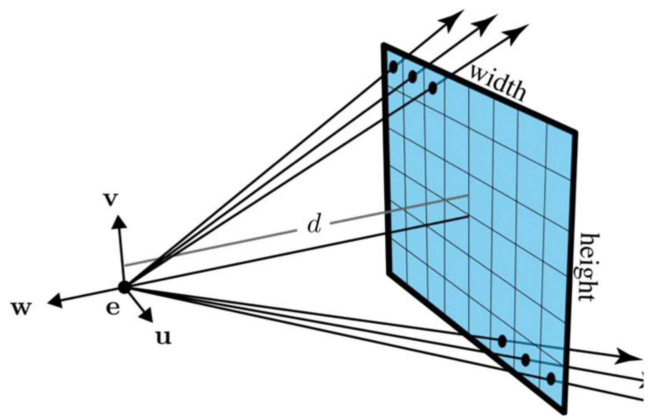
Bottom Left Corner (i, j) : $(-\frac{1}{2}, -\frac{1}{2})$

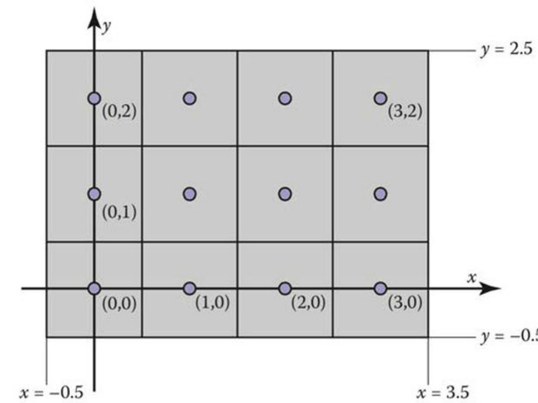
Top Right Corner (i, j) : $(n_x - \frac{1}{2}, n_y - \frac{1}{2})$

Bottom Left Corner (u, v) : $(-\frac{\text{width}}{2}, -\frac{\text{height}}{2})$

Top Right Corner (u, v) : $(\frac{\text{width}}{2}, \frac{\text{height}}{2})$

pixel at position (i, j) in the raster image has the position:





Physical Width of Image in 3D Space

Num Pixels X

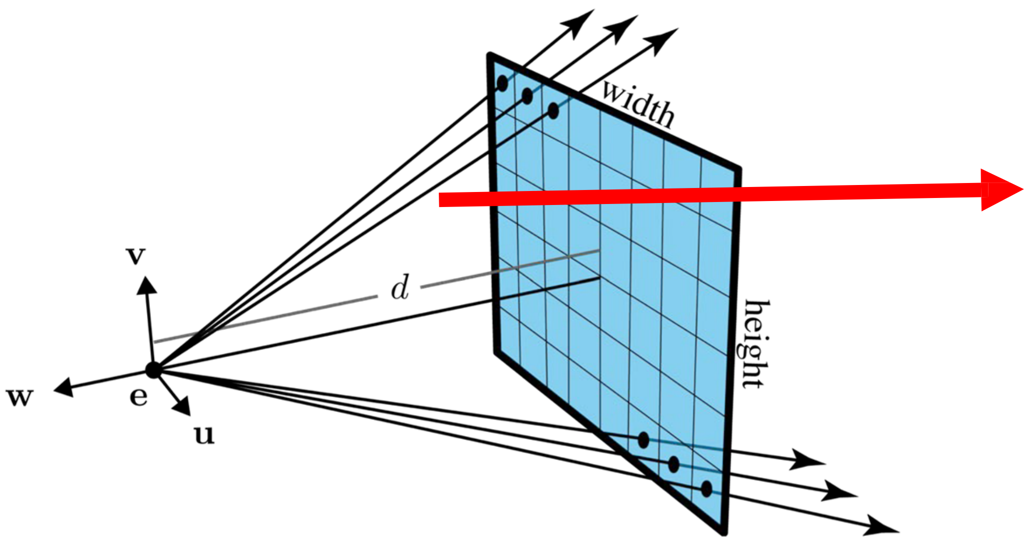
$$u = \frac{\text{width}}{n_x} \cdot \left(i + \frac{1}{2}\right) - \frac{\text{width}}{2}$$

Physical Height of Image in 3D Space

Num Pixels Y

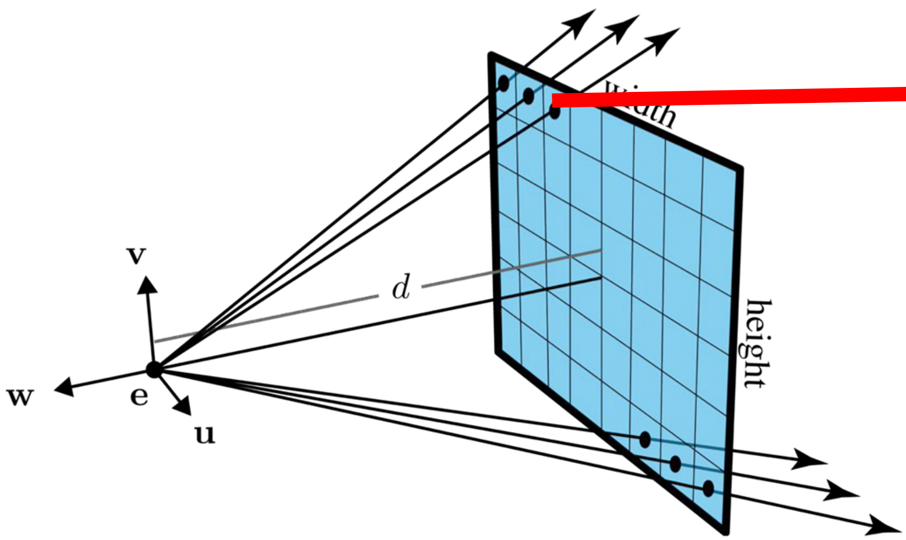
$$v = \frac{\text{height}}{n_y} \cdot \left(j + \frac{1}{2}\right) - \frac{\text{height}}{2}$$

Ray Equation in Camera Space



$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

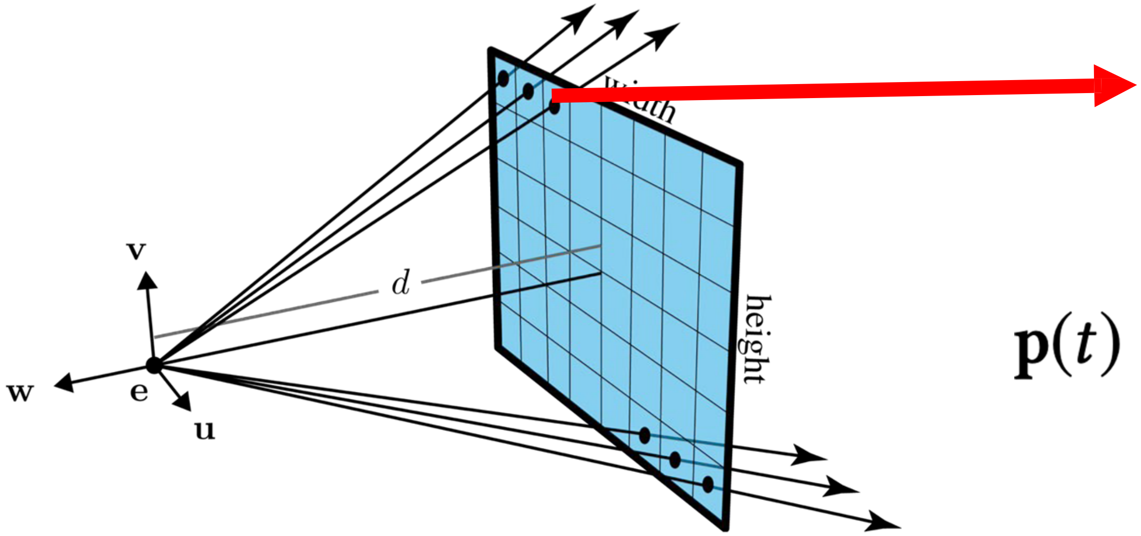
Ray Equation in Camera Space



$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

$$\mathbf{p}(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \left(\begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right)$$

Ray Equation in Camera Space



$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

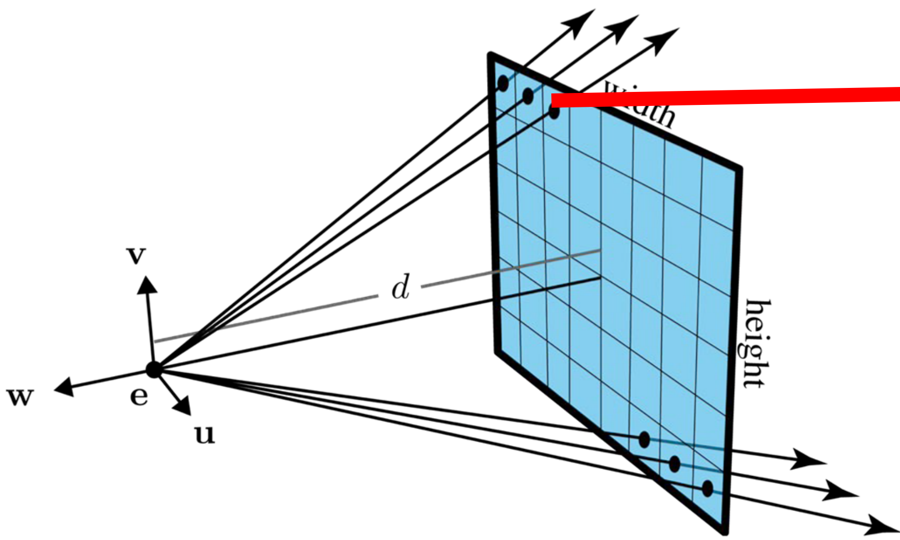
$$\mathbf{p}(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t\left(\begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right)$$

$$\mathbf{p}(t) = t\begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix}$$

$$u = \frac{\text{width}}{n_x} \cdot \left(i + \frac{1}{2}\right) - \frac{\text{width}}{2}$$

$$v = \frac{\text{height}}{n_y} \cdot \left(j + \frac{1}{2}\right) - \frac{\text{height}}{2}$$

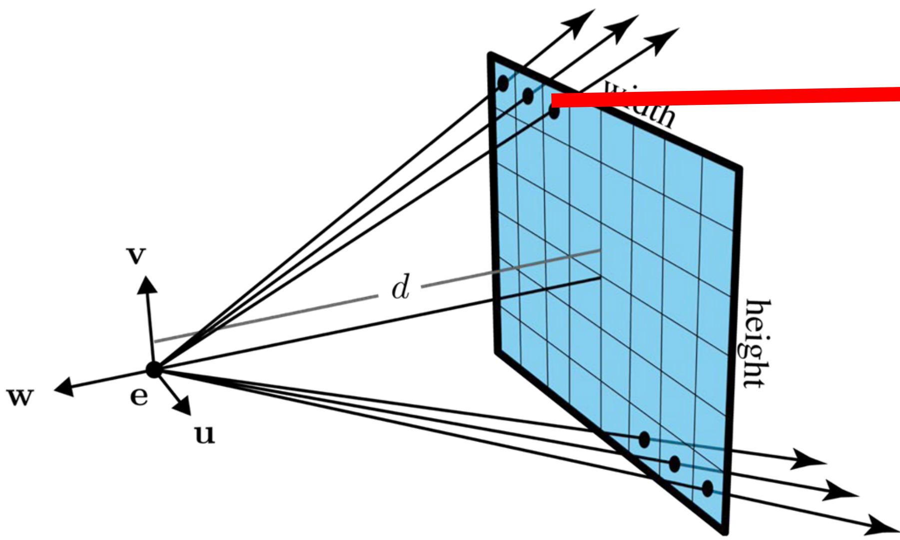
Ray Equation in World Space



$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

$$\mathbf{p}(t) = \mathbf{e} + t\,(u(i)\mathbf{u} + v(j)\mathbf{v} + -d\mathbf{w})$$

Ray Equation in World Space



$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

$$\mathbf{p}(t) = \mathbf{e} + t \begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} \end{bmatrix} \begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix}$$

Camera Transformation Matrix

Ray Casting

```
for each pixel in the image {
    Generate a ray
    for each object in the scene {
        if (Intersect ray with object){
            Set pixel colour
        }
    }
}
```

Example – Pixel Space

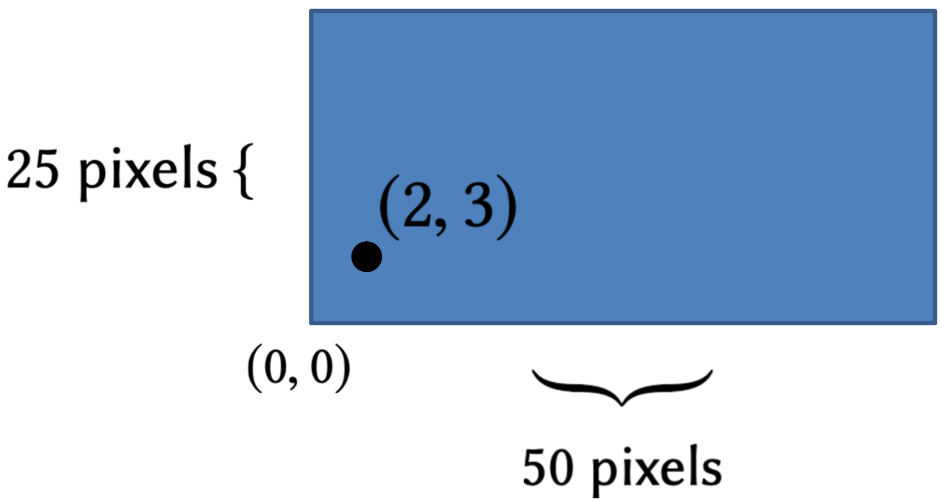
$$\text{width} = 2$$

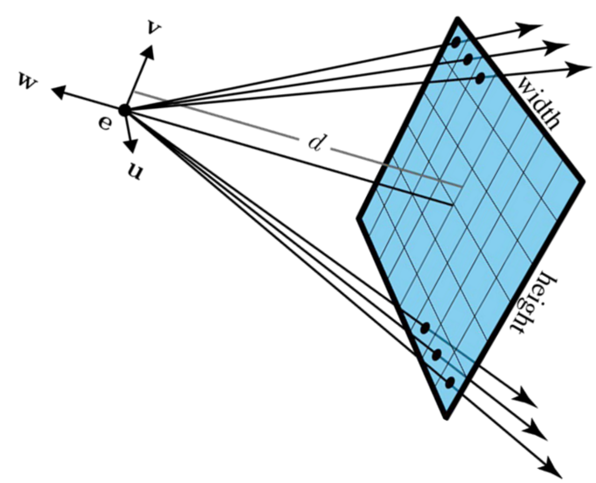
$$\text{height} = 1$$

$$n_x = 50$$

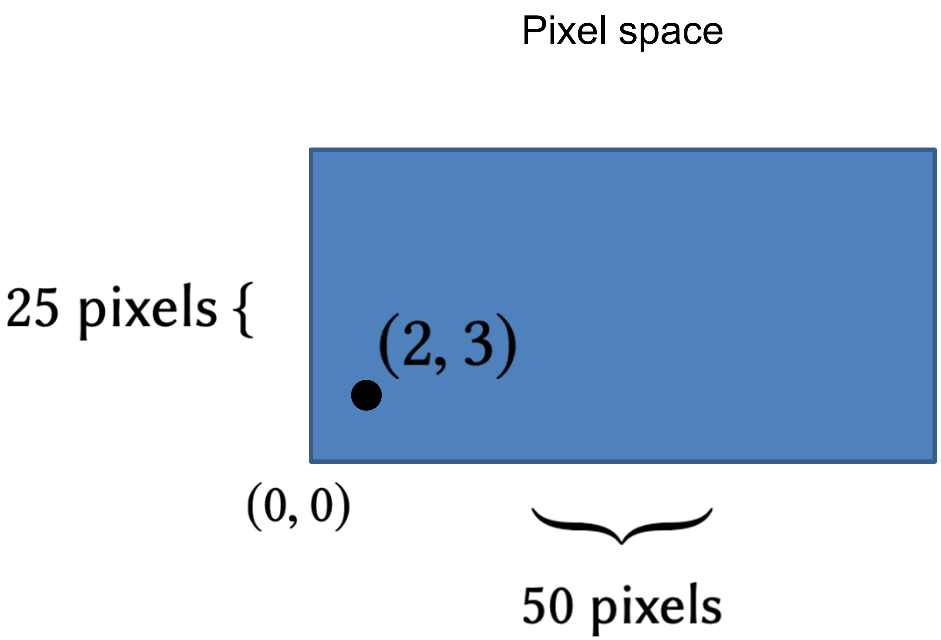
$$n_y = 25$$

Pixel space





Example – Pixel Space



$$\text{width} = 2$$

$$\text{height} = 1$$

$$n_x = 50$$

$$n_y = 25$$

$$u = \frac{\text{width}}{n_x} \cdot \left(i + \frac{1}{2}\right) - \frac{\text{width}}{2}$$

$$v = \frac{\text{height}}{n_y} \cdot \left(j + \frac{1}{2}\right) - \frac{\text{height}}{2}$$

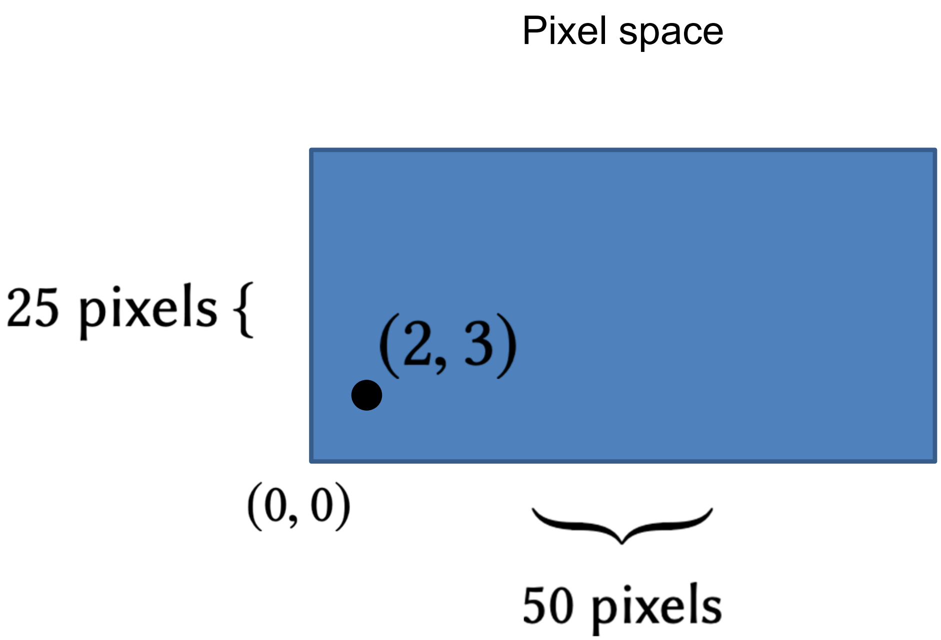
Example – Pixel Space

$$\text{width} = 2$$

$$\text{height} = 1$$

$$n_x = 50$$

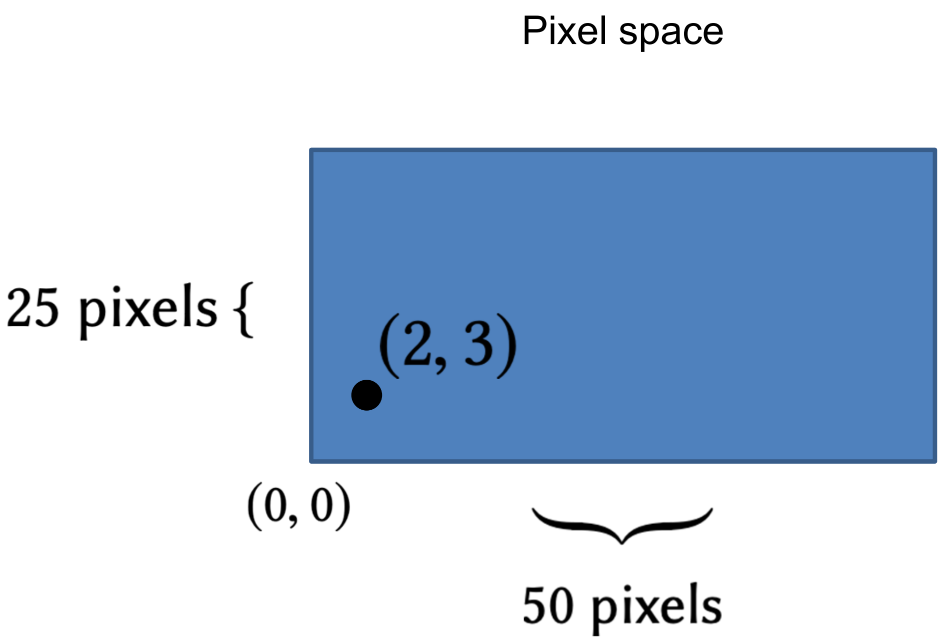
$$n_y = 25$$



$$u = \frac{2}{50} \cdot \left(2 + \frac{1}{2}\right) - \frac{2}{2}$$

$$v = \frac{1}{25} \cdot \left(3 + \frac{1}{2}\right) - \frac{1}{2}$$

Example – Pixel Space



$$\text{width} = 2$$

$$\text{height} = 1$$

$$n_x = 50$$

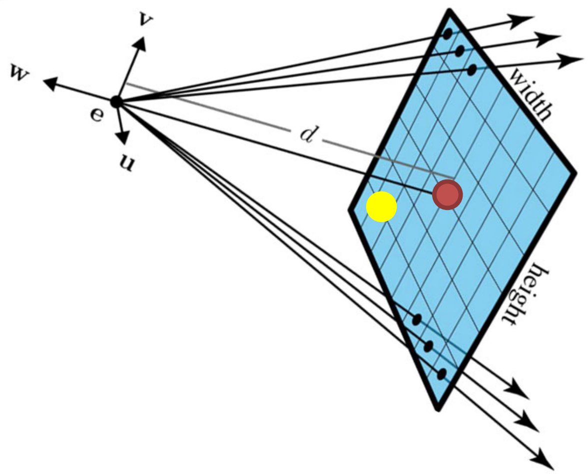
$$n_y = 25$$

$$u = -0.9$$

$$v = -0.36$$

Example – Pixel Space

Camera space



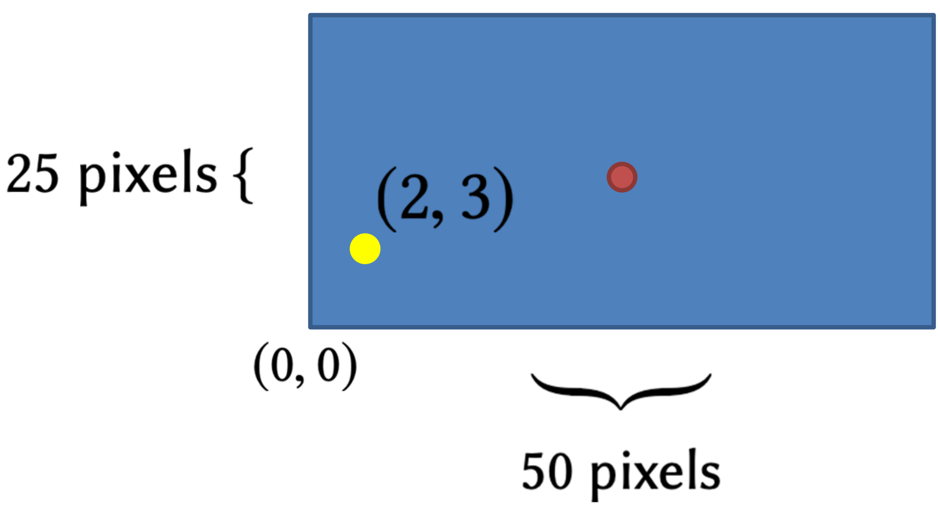
$$width = 2$$

$$height = 1$$

$$n_x = 50$$

$$n_y = 25$$

Pixel space

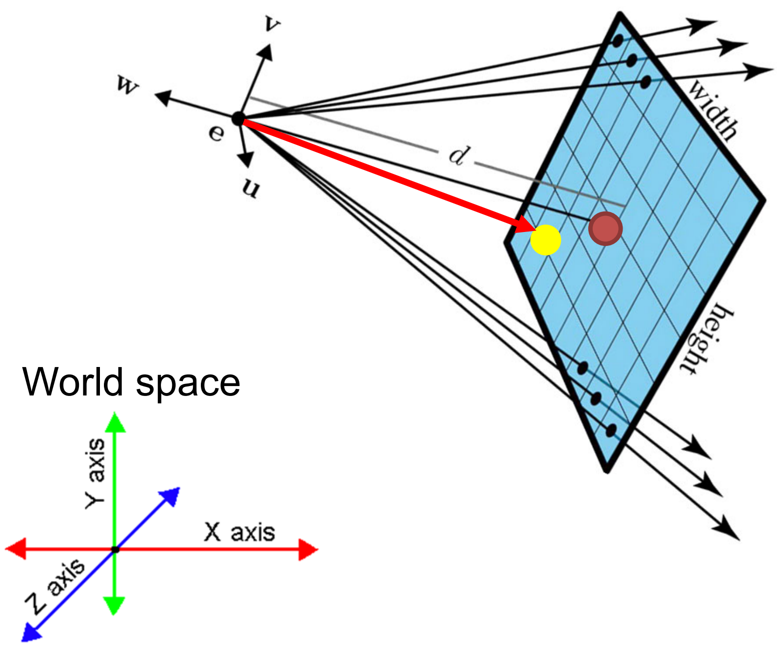


$$u = -0.9$$

$$v = -0.36$$

Example – Camera Space

Camera space



$$\mathbf{u}, \mathbf{v}, \mathbf{w}$$

are the basis vectors for camera space

$$d = 10$$

d is the distance from the viewpoint to the image plane (focal length)

$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

our ray equation

Example – Camera Space

$d = 10$

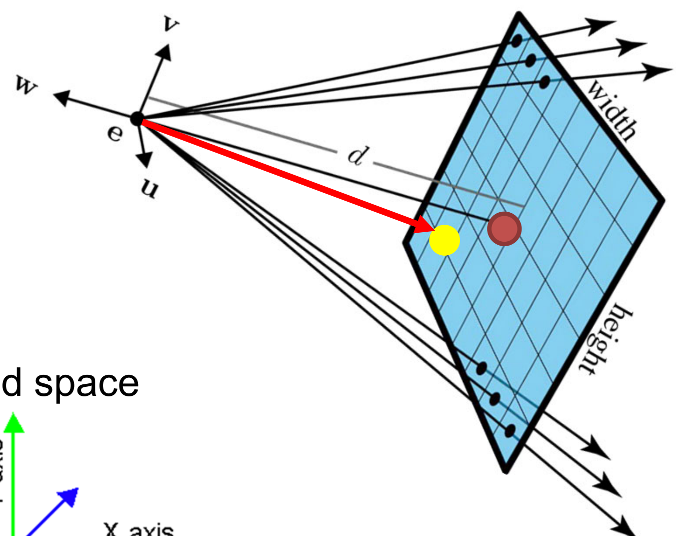
$$\mathbf{u}, \mathbf{v}, \mathbf{w}$$

are the basis vectors for camera space

$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

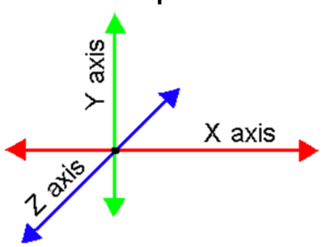
our ray equation

Camera space



$$\mathbf{p}(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t\left(\begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right)$$

World space



$d = 10$

Example – Camera Space

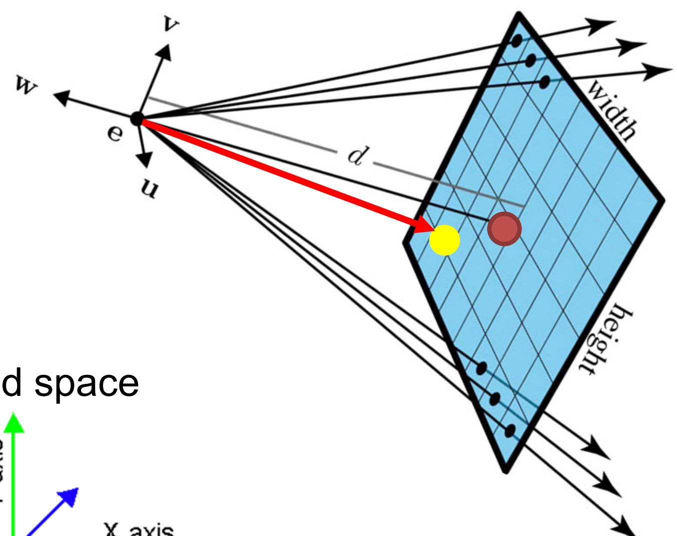
$$\mathbf{u}, \mathbf{v}, \mathbf{w}$$

are the basis vectors for camera space

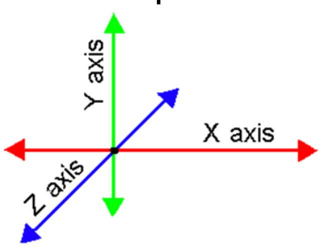
$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

our ray equation

Camera space



World space



for (i,j) = (2,3)

$$u(2) = -0.9, v(3) = -0.36$$

$$\mathbf{p}(t) = t \begin{bmatrix} -0.9 \\ -0.36 \\ -10 \end{bmatrix}$$

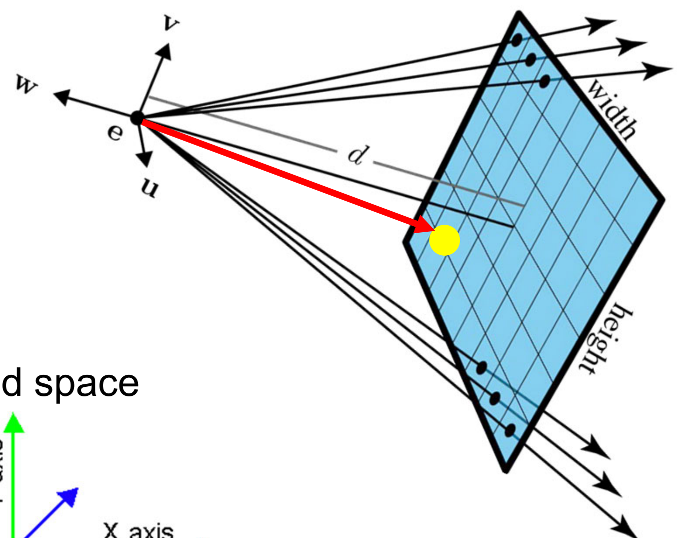
Example – World Space

$$\mathbf{u}, \mathbf{v}, \mathbf{w}$$

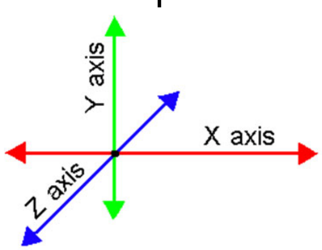
are the basis vectors for camera space

$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

our ray equation



World space



$$\mathbf{e} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\mathbf{p}(t) = \mathbf{e} + t\left(u(i)\mathbf{u} + v(j)\mathbf{v} + -d\mathbf{w}\right)$$

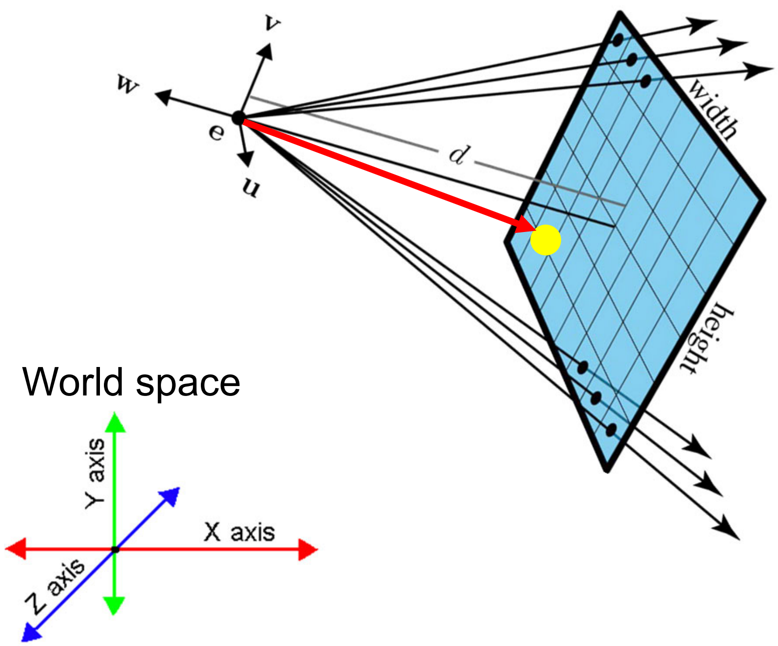
Example – World Space

$$\mathbf{u}, \mathbf{v}, \mathbf{w}$$

are the basis vectors for camera space

$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

our ray equation



$$\mathbf{e} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\mathbf{p}(t) = \mathbf{e} + t \begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} \end{bmatrix} \begin{bmatrix} u(i) \\ v(j) \\ -d \end{bmatrix}$$

Example – World Space

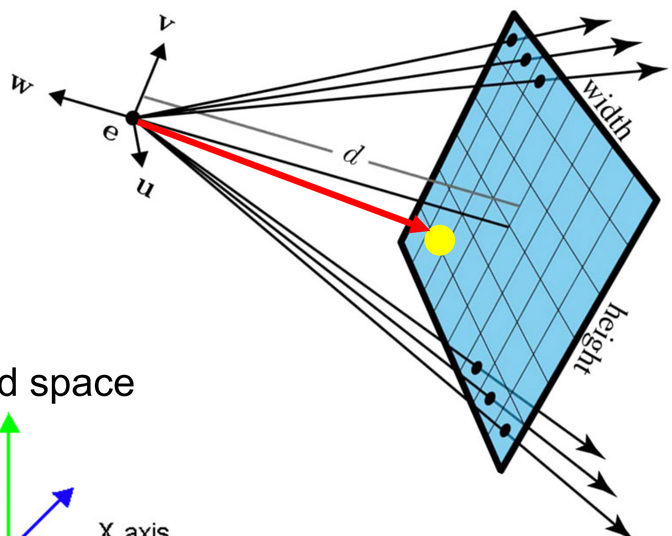
$$\mathbf{e} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\mathbf{u}, \mathbf{v}, \mathbf{w}$$

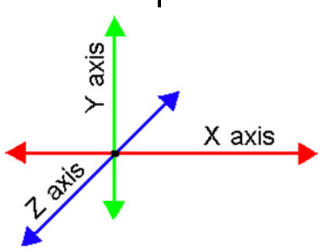
are the basis vectors for camera space

$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

our ray equation



World space



$$\mathbf{p}(t) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -0.9 \\ -0.36 \\ -10 \end{bmatrix}$$

https://mathworld.wolfram.com/ChangeofCoordinatesMatrix.html

Ray Casting

Some Slides/Images adapted from Marschner and Shirley and David Levin

Basic Components of Ray Casting

Ray

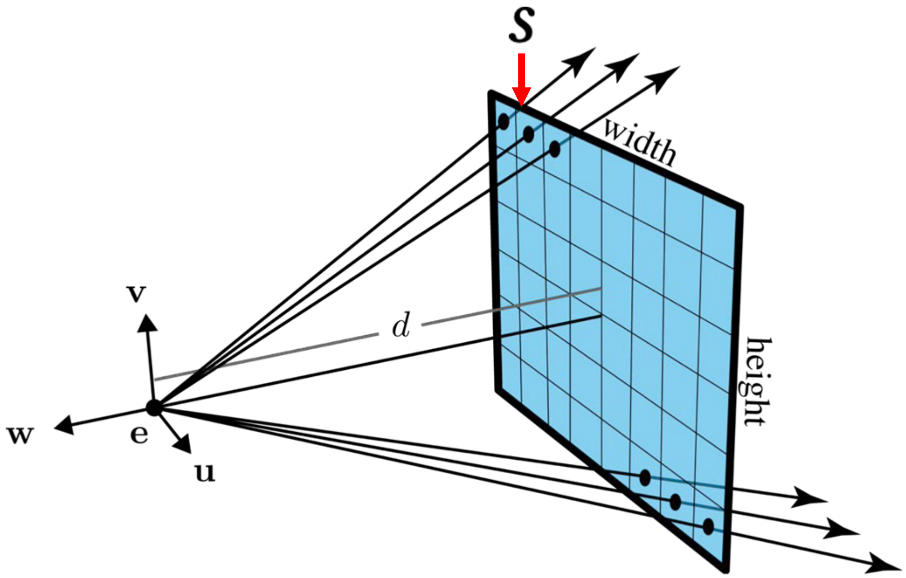
Camera

Intersection Tests

Intersection Tests

- Plane
- Sphere
- Triangle

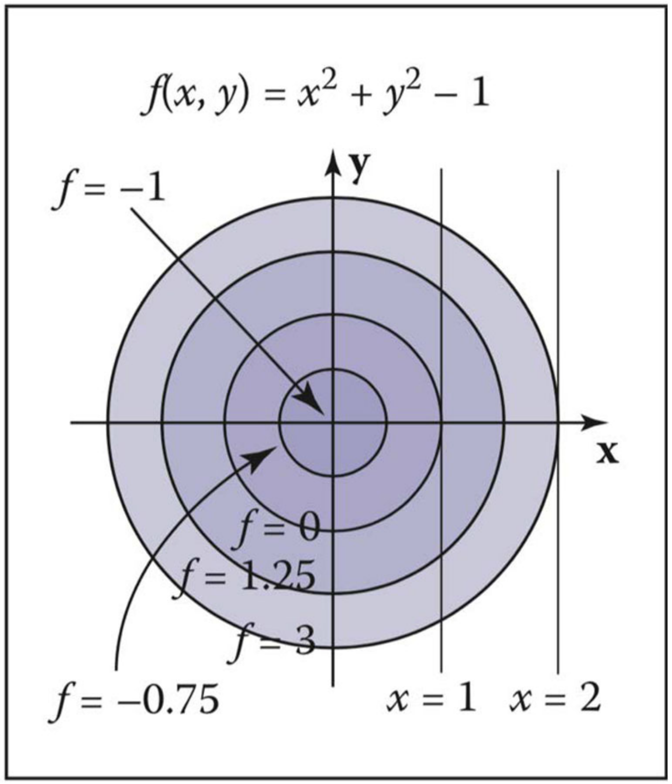
Ray Equation



$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

Aside: Types of Surface

Implicit Surface

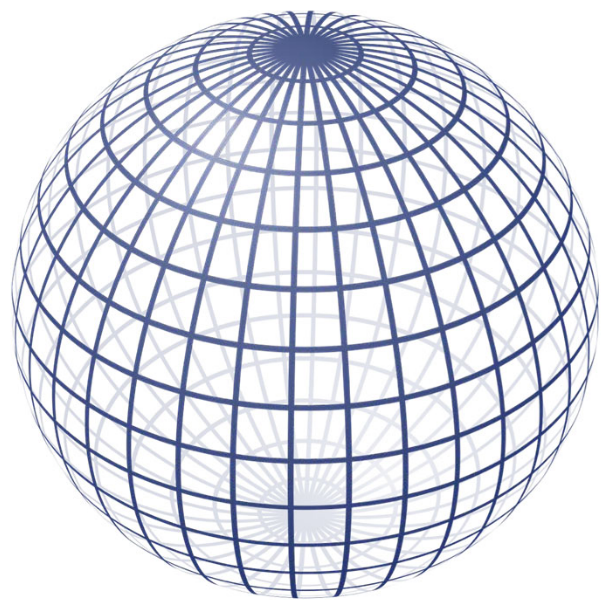


Parametric Surface

$$x = r\cos\phi\sin\theta,$$

$$y = r\sin\phi\sin\theta,$$

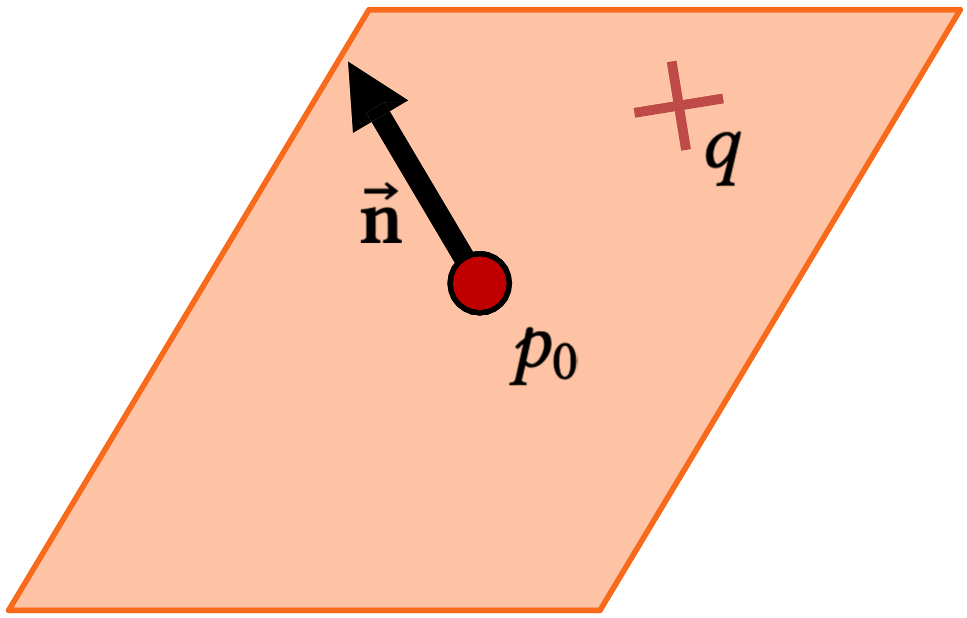
$$z = r\cos\theta.$$



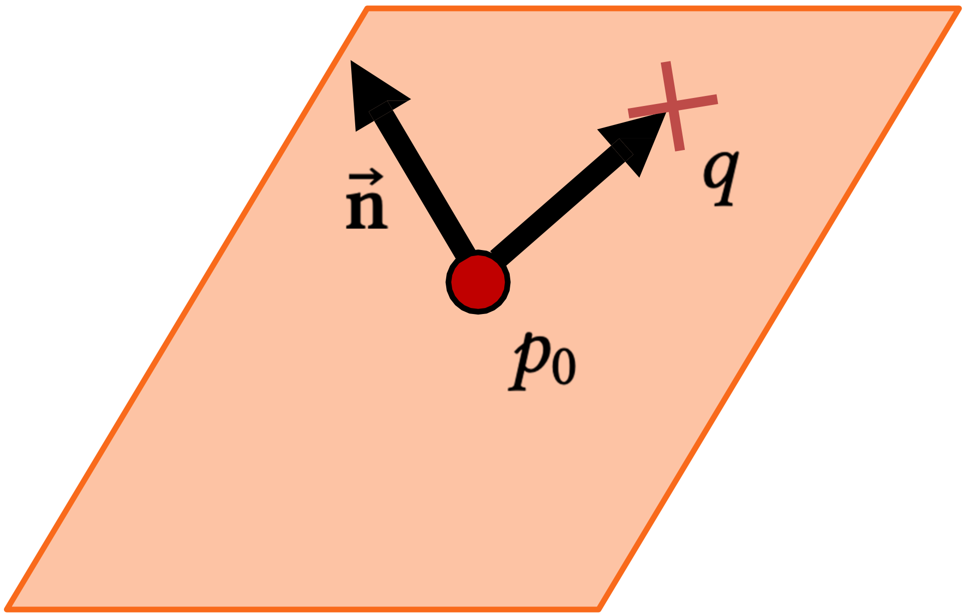
Intersection Tests

- Plane
- Sphere
- Triangle

Plane Equation

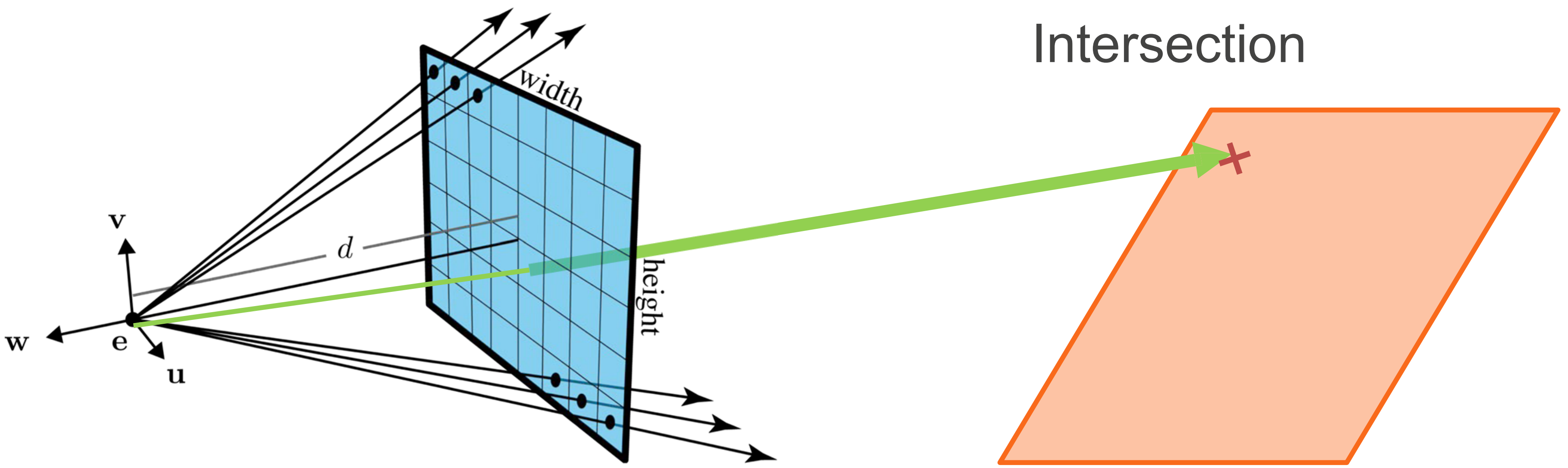


Plane Equation



$$\vec{\mathbf{n}} \cdot (q - p_0) = 0$$

Ray-Plane Intersection

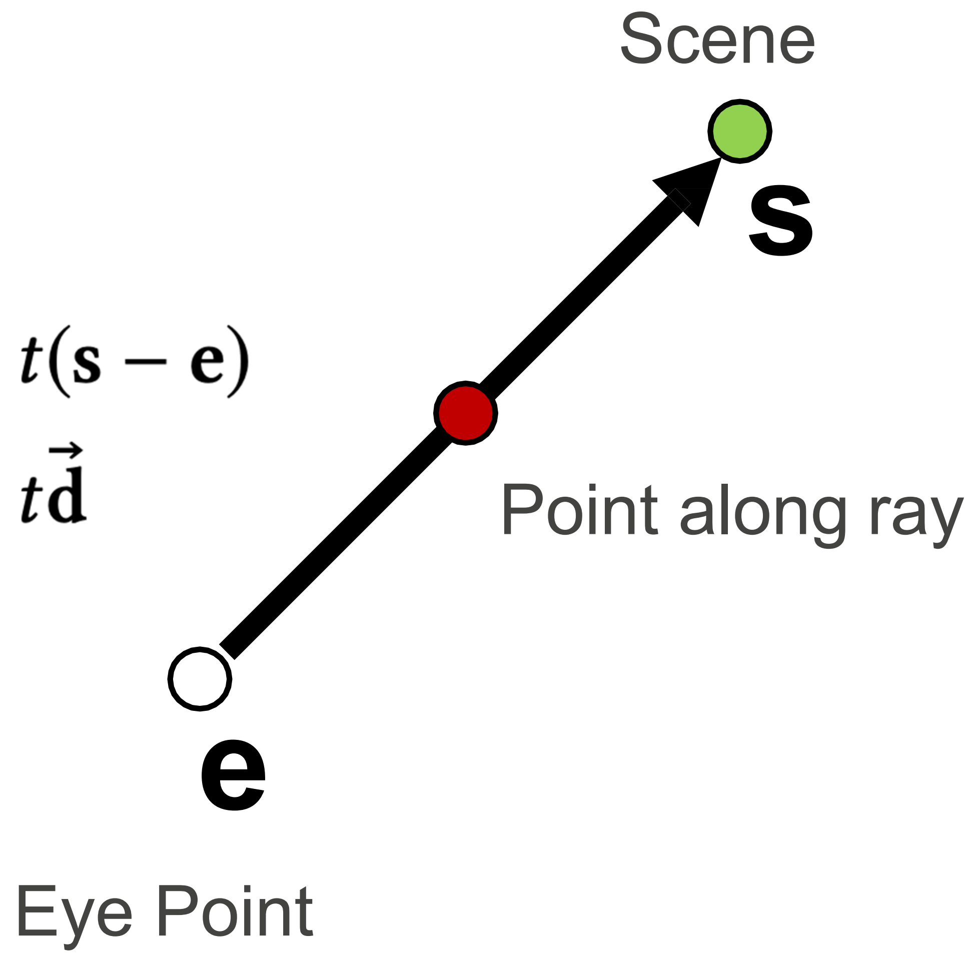


Ray-Plane Intersection

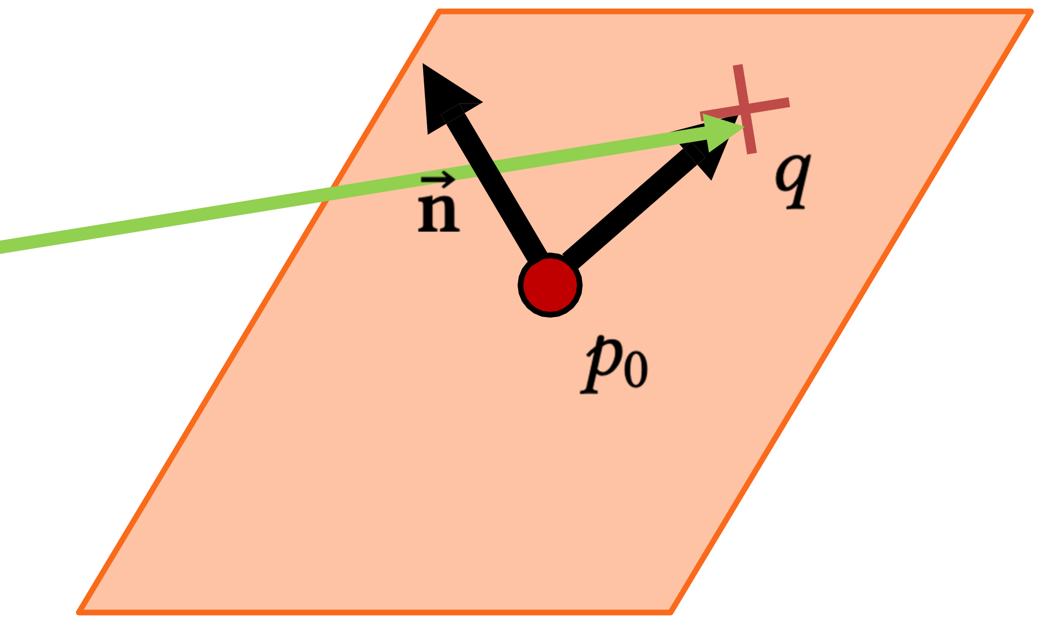
Ray equation

$$\mathbf{p}(t) = \mathbf{e} + t(\mathbf{s} - \mathbf{e})$$

$$\mathbf{p}(t) = \mathbf{e} + t\vec{\mathbf{d}}$$



Plane Equation

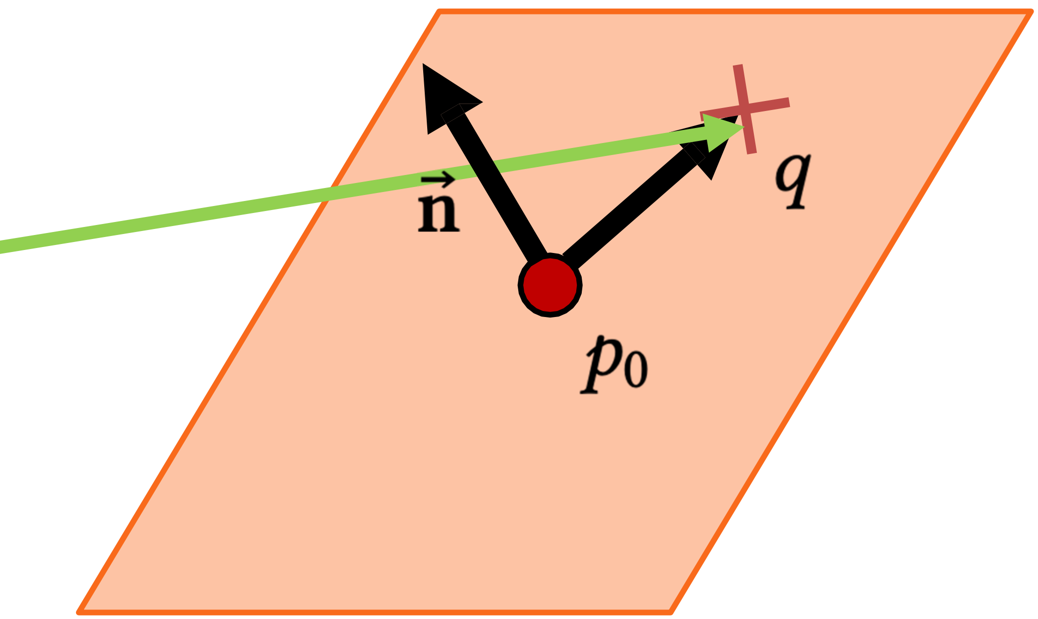


Plane equation
Substitute ray equation into it

$$\vec{\mathbf{n}} \cdot (\mathbf{p}(t) - p_0) = 0$$

$$\vec{\mathbf{n}} \cdot ((\mathbf{e} + t\vec{\mathbf{d}}) - p_0) = 0$$

Plane Equation



Plane equation
Substitute ray equation into it

$$\vec{\mathbf{n}} \cdot (\mathbf{p}(t) - p_0) = 0$$

$$\vec{\mathbf{n}} \cdot ((\mathbf{e} + t\vec{\mathbf{d}}) - p_0) = 0$$

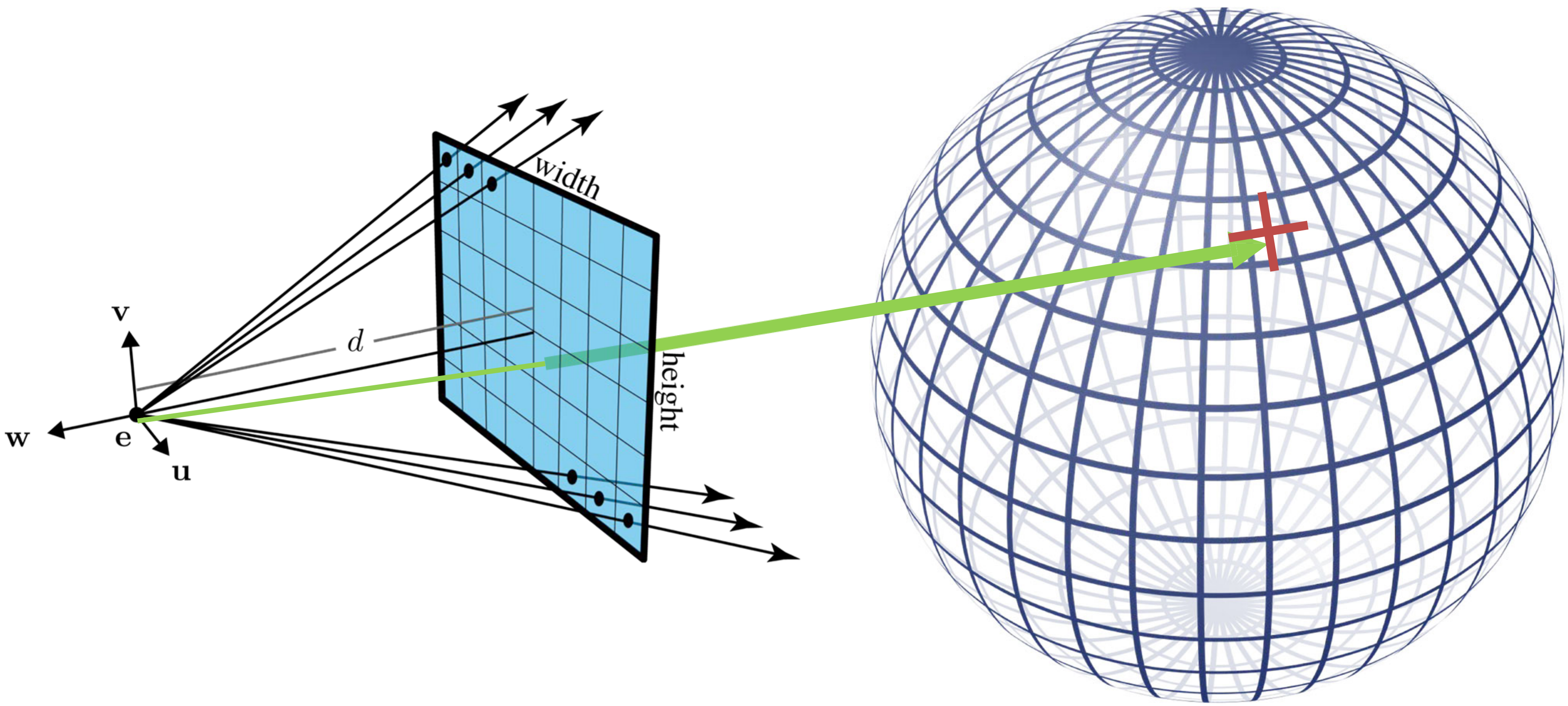
Solve for t

$$\boxed{t = \frac{-\vec{\mathbf{n}} \cdot (\mathbf{e} - p_0)}{\vec{\mathbf{n}} \cdot \vec{\mathbf{d}}}}$$

Intersection Tests

- Plane
- Sphere
- Triangle

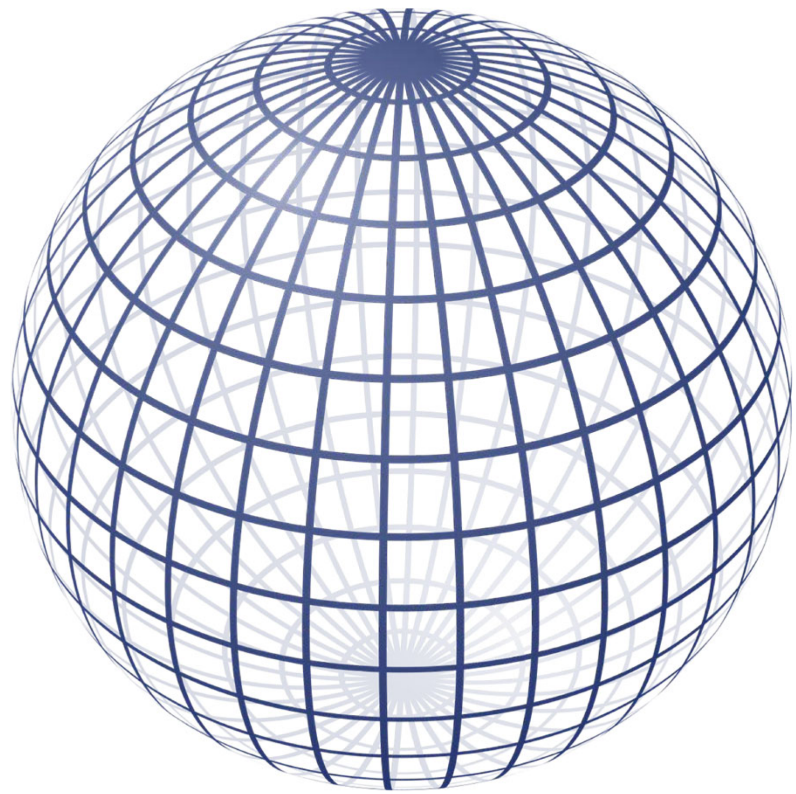
Ray-Sphere Intersection



Implicit Equation of a Sphere

$$(q - \mathbf{c}) \cdot (q - \mathbf{c}) - r^2 = 0$$

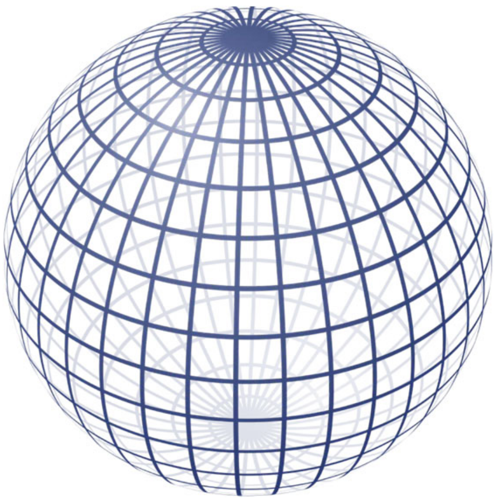
*Sphere centered at **c** with radius r*



Ray-Sphere Intersection

Substitute ray equation into implicit equation for sphere

$$(\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) \cdot (\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) - r^2 = 0$$



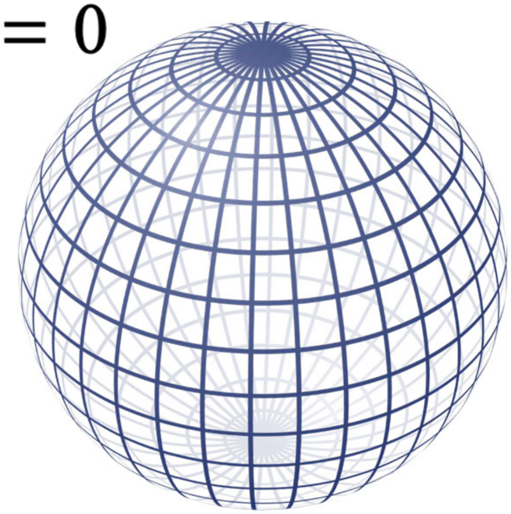
Ray-Sphere Intersection

Substitute ray equation into implicit equation for sphere

$$(\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) \cdot (\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) - r^2 = 0$$

Rearrange

$$(\vec{\mathbf{d}} \cdot \vec{\mathbf{d}})t^2 + 2\vec{\mathbf{d}} \cdot (\mathbf{e} - \mathbf{c})t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - r^2 = 0$$



Ray-Sphere Intersection

Substitute ray equation into implicit equation for sphere

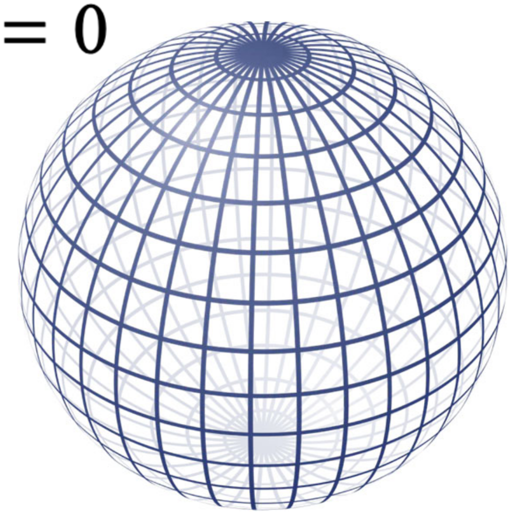
$$(\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) \cdot (\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) - r^2 = 0$$

Rearrange

$$(\vec{\mathbf{d}} \cdot \vec{\mathbf{d}})t^2 + 2\vec{\mathbf{d}} \cdot (\mathbf{e} - \mathbf{c})t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - r^2 = 0$$

Looks familiar…

$$At^2 + Bt + C = 0$$



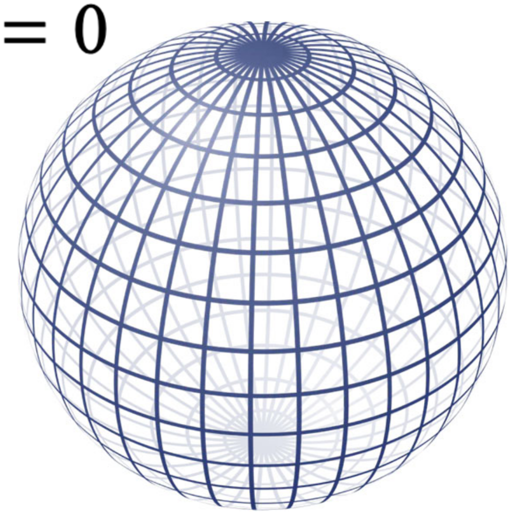
Ray-Sphere Intersection

Substitute ray equation into implicit equation for sphere

$$(\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) \cdot (\mathbf{e} + t\vec{\mathbf{d}} - \mathbf{c}) - r^2 = 0$$

Rearrange

$$(\vec{\mathbf{d}} \cdot \vec{\mathbf{d}})t^2 + 2\vec{\mathbf{d}} \cdot (\mathbf{e} - \mathbf{c})t + (\mathbf{e} - \mathbf{c}) \cdot (\mathbf{e} - \mathbf{c}) - r^2 = 0$$



Looks familiar…

$$At^2 + Bt + C = 0$$

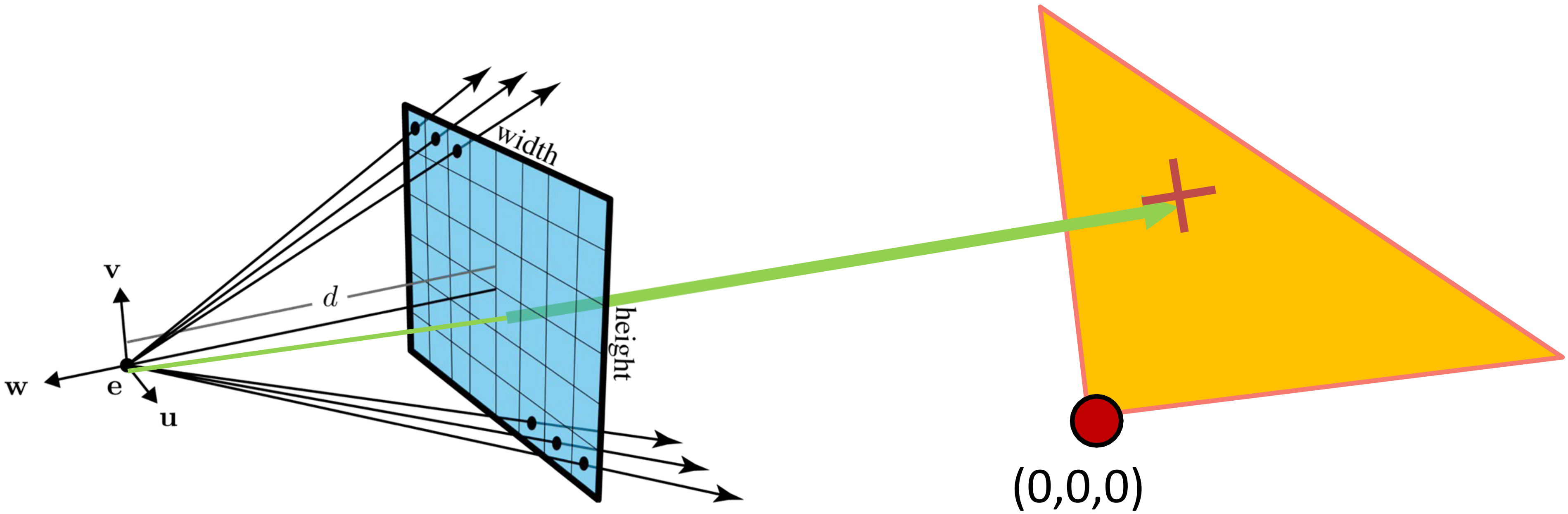
It’s a quadratic! (can use the quadratic equation)

Hint: the discriminant tells us what kinds of roots the equation has.

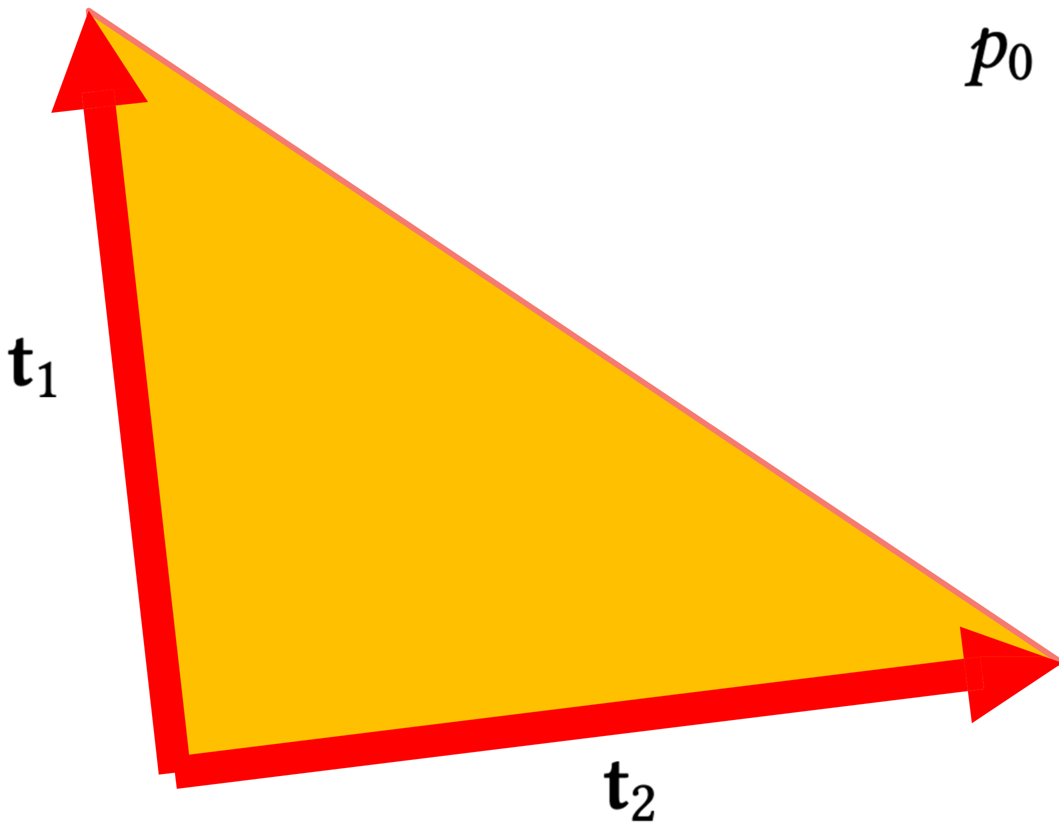
Intersection Tests

- Plane
- Sphere
- Triangle

Ray-Triangle Intersection

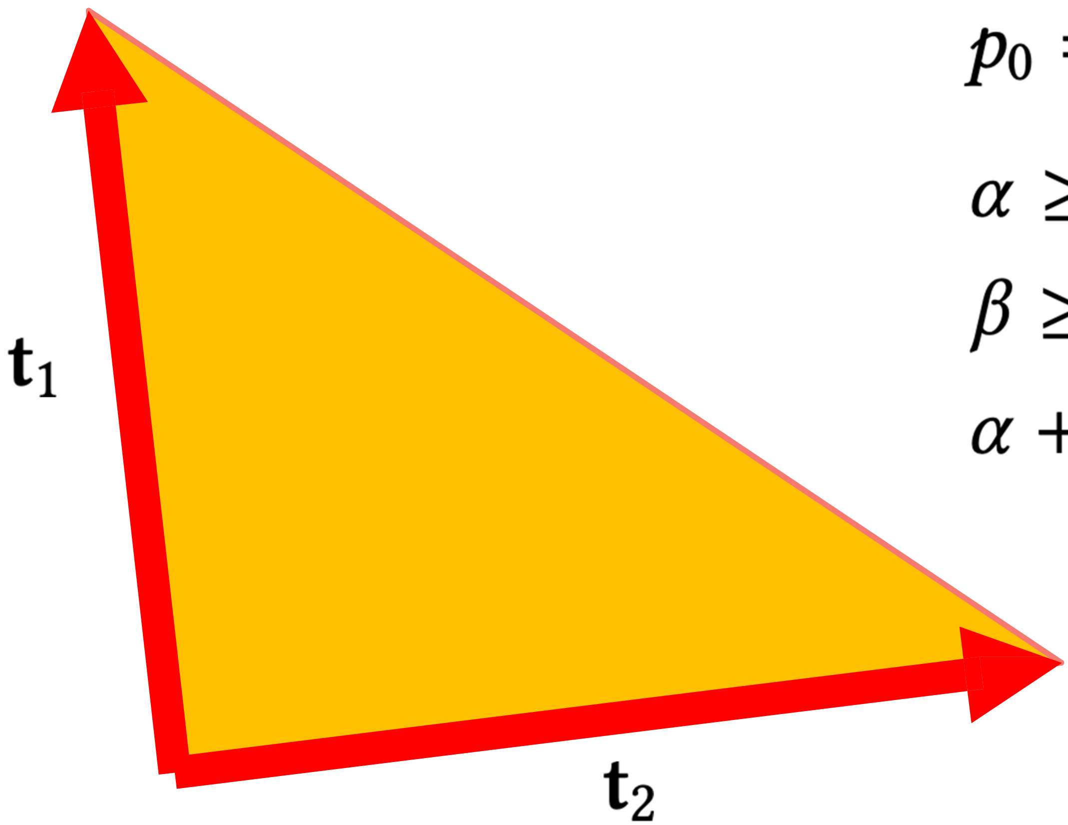


Equations for a Triangle



$$p_0 = \alpha\mathbf{t}_1 + \beta\mathbf{t}_2$$

Equations for a Triangle



$$p_0 = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2$$

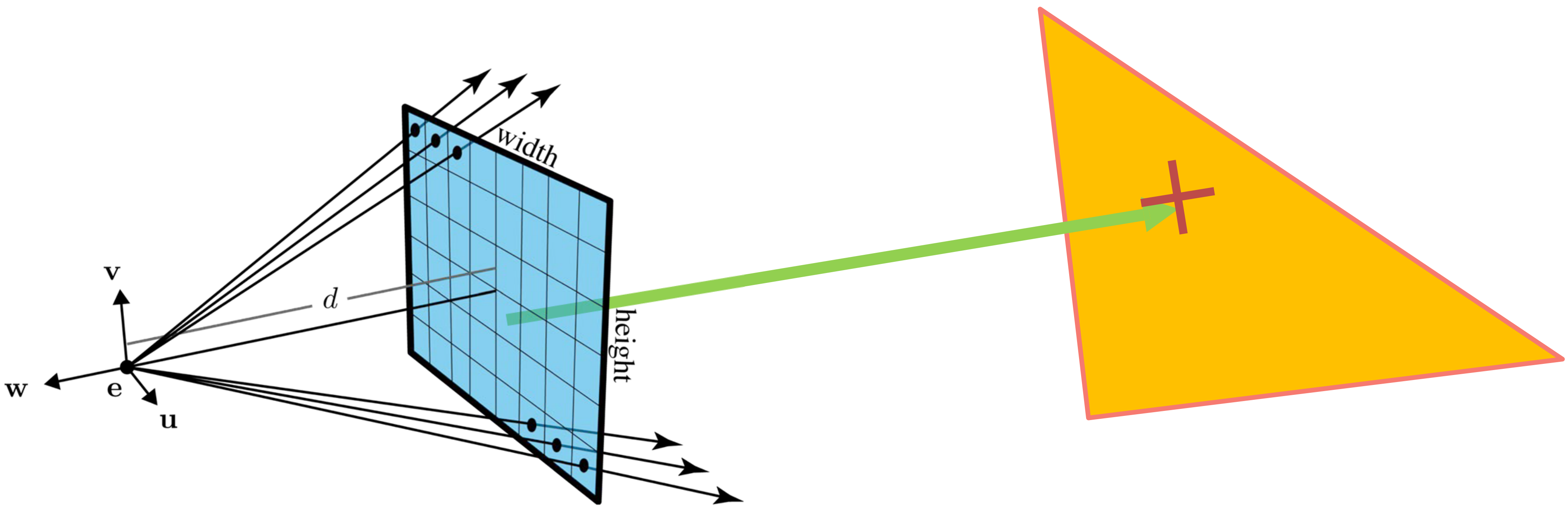
$$\alpha \geq 0$$

$$\beta \geq 0$$

$$\alpha + \beta \leq 1$$

Intersection with a Triangle (Parametric Surface)

Check via equating point on surface with point on ray



Intersection with a Triangle (Parametric Surface)

Check via equating point on surface with point on ray

$$\mathbf{p}(t) = \alpha\mathbf{t}_1 + \beta\mathbf{t}_2$$

$$\mathbf{e} + t\vec{\mathbf{d}} = \alpha\mathbf{t}_1 + \beta\mathbf{t}_2$$

$$\mathbf{e} = \alpha\mathbf{t}_1 + \beta\mathbf{t}_2 - t\vec{\mathbf{d}}$$

Intersection with a Triangle (Parametric Surface)

Check via equating point on surface with point on ray

$$\mathbf{e} = \alpha \mathbf{t}_1 + \beta \mathbf{t}_2 - t\vec{\mathbf{d}}$$

$$\mathbf{e} = \begin{bmatrix} \mathbf{t}_1 & \mathbf{t}_2 & -\mathbf{d} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ t \end{bmatrix}$$

Check values of α, β, t

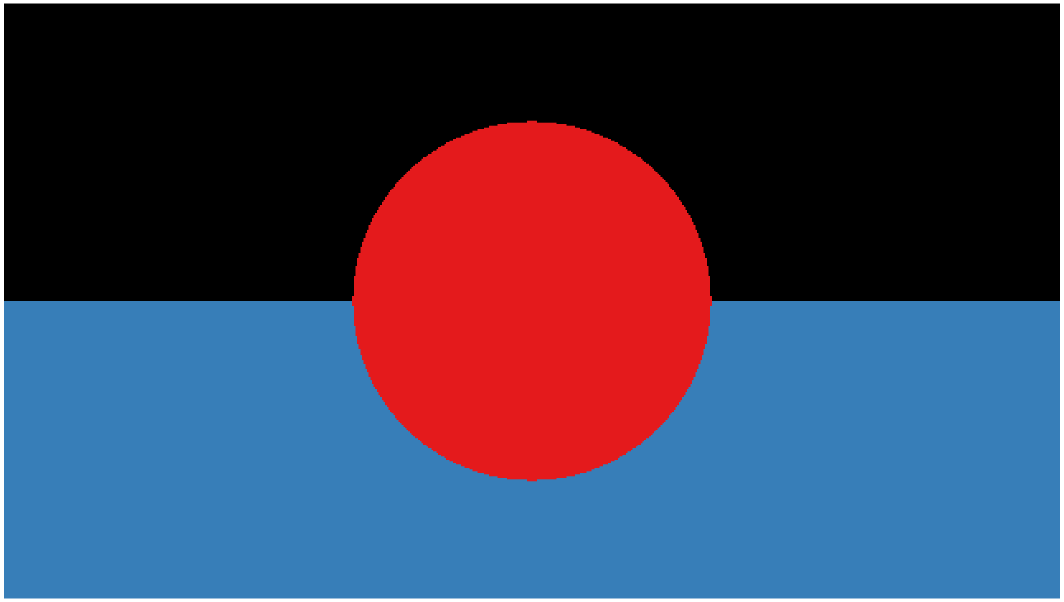
Ray Casting

```
for each pixel in the image {
     Generate a ray

     for each object in the scene {

          if (Intersect ray with object)

               {  Set pixel colour

          }

     }
}
```

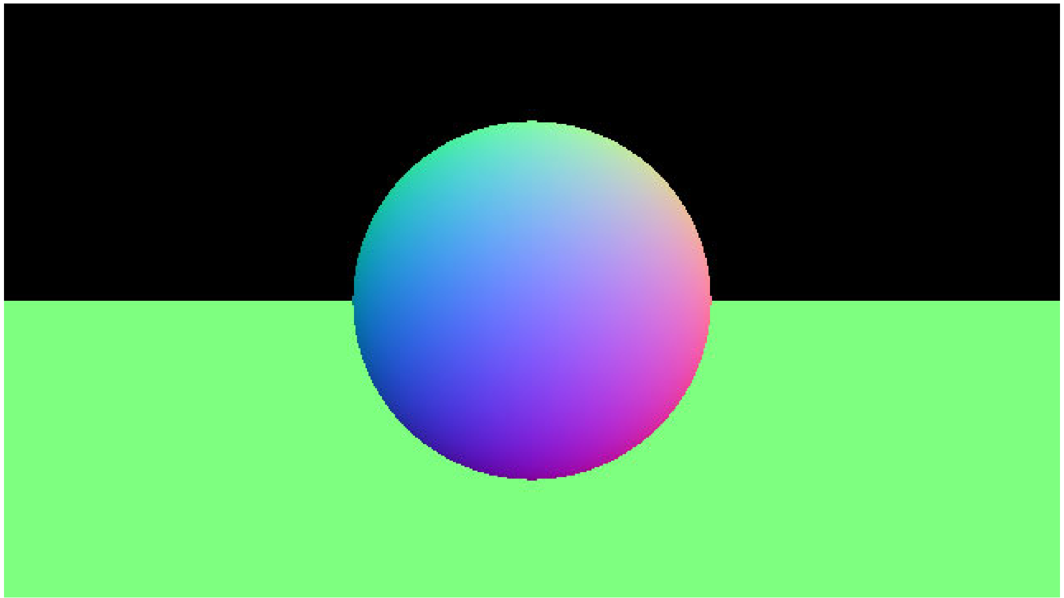
Output Type

- Object ID
- Surface Normal
- Depth

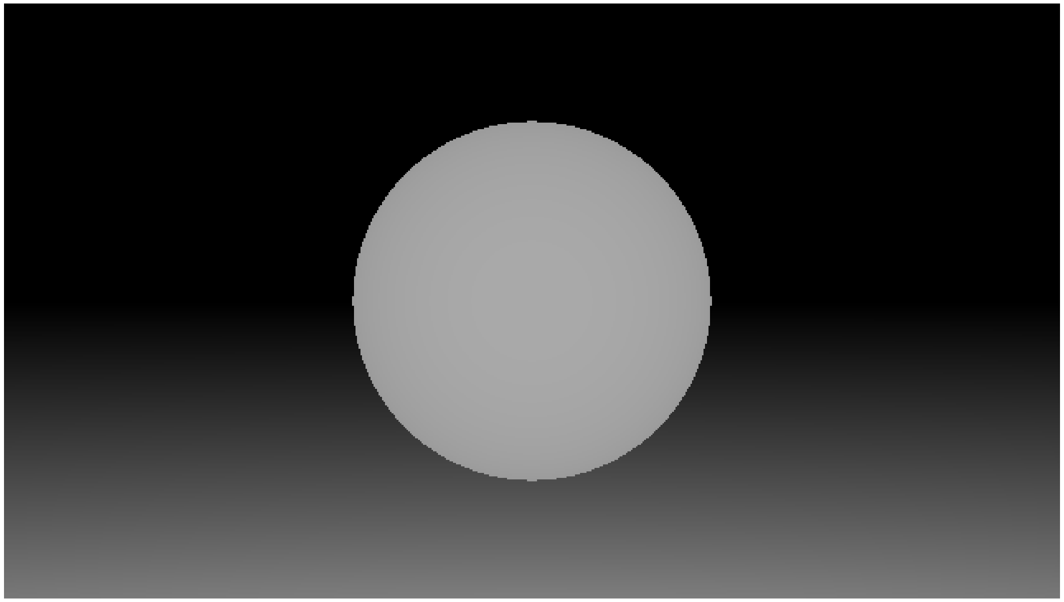
Output Type: Object ID



Output Type: Surface Normal



Output Type: Depth



Assignment #2 tasks

`src/viewing_ray.cpp`

Construct a viewing ray given a camera and subscripts to a pixel.

`src/first_hit.cpp`

Find the first (visible) hit given a ray and a collection of scene objects

`Sphere::intersect_ray` in `src/Sphere.cpp`

Intersect a sphere with a ray.

`Plane::intersect_ray` in `src/Plane.cpp`

Intersect a plane with a ray.

`Triangle::intersect_ray` in `src/Triangle.cpp`

Intersect a triangle with a ray.

`TriangleSoup::intersect_ray` in `src/TriangleSoup.cpp`

Intersect a triangle soup with a ray.

Next Week

Ray Tracing